Journal of Strategic Information Systems 33 (2024) 101863

Contents lists available at ScienceDirect

Journal of Strategic Information Systems

journal homepage: www.elsevier.com/locate/jsis

Time to reassess data value: The many faces of data in organizations

Daisy Xu [a,*], Marta Indulska [b], Ida Asadi Someh [b], Graeme Shanks [b]

[a] ARC Centre for Information Resilience (CIRES), The University of Queensland, Australia
[b] Business School, The University of Queensland, Australia

ABSTRACT

Despite the substantial body of evidence detailing the multifaceted use of data within organizations, the conceptualizations of data and their value propositions remain disjointed and require updating. Information Systems scholars contend that the traditional ways of conceiving data now appear inadequate in framing this ever-evolving data-driven phenomenon. In this context, we argue for a reassessment of the fundamental assumptions about data in the field. This paper offers a comprehensive literature review, through which we conceptualize the role of data into four distinguishable types: data as a tool, as a commodity, as a practice, and as algorithmic intelligence. Each type possesses a set of identifiable characteristics, usage, and unique pathways of value creation. Together these elements form a typology, which provides an explanation for the intricate and complex nature of data use in organizations and the diverse sources of their value.

Introduction

In today's business landscape, everything is data — not only the static, structured data collected from customers, employees, operations, and transactions, but also the continuously updating, unstructured data from social media, enterprise systems, and other platforms (Baesens et al., 2014; Constantiou and Kallinikos, 2015). Additionally, data may also include the digital imprints that users leave when interacting with devices, facilities, people, and organizations in their everyday lives (Gregory et al., 2021). These diverse forms and sources of data can imply substantial value by addressing specific organizational objectives (Abbasi et al., 2016; Grover et al., 2018).

While significant progress has been made by Information Systems (IS) researchers in recognizing the strategic importance of data for organizations (Brynjolfsson and McElheran, 2016; Chen et al., 2012; Grover et al., 2018; Hagiu and Wright, 2020), robust evidence has shown that the ways in which data contribute to business have evolved over time and grown increasingly varied (Aaltonen et al., 2023; Xu et al., 2023). Data can be used to inform and improve managerial decision-making (Brynjolfsson and McElheran, 2016; Ghasemaghaei, 2019). Data can also be embedded in organizational processes to shape digital innovations in products, process, and business model (Hagiu and Wright, 2020; Gregory et al., 2021; Grover et al., 2018). Additionally, data can be shared across organizations, or traded to generate direct revenue (Aaltonen et al., 2021). Data can even simulate human intelligence to automate business processes, and prescribe programmed actions without human interventions (Shollo et al., 2022). We argue that the diverse usage of data in organizations reveals the changing nature and multiple pathways of value creation, which needs to be addressed and managed accordingly.

In our review of the literature, we observe a pressing challenge in the scholarly realm, specifically, in the ways we conceive of data

* Corresponding author.
E-mail addresses: daisy.xu@uq.edu.au (D. Xu), m.indulska@business.uq.edu.au (M. Indulska), i.asadi@business.uq.edu.au (I. Asadi Someh), g.shanks@business.uq.edu.au (G. Shanks).

https://doi.org/10.1016/j.jsis.2024.101863
Received 1 February 2023; Received in revised form 14 September 2024; Accepted 14 September 2024
Available online 9 October 2024
0963-8687/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

and their value. First, there has been limited explorations of data regarding their nature, effects, and character, upon which the data phenomena are based (Zins, 2007; Jones, 2019). Up until today, conceptualizations of data are still found to be "bedevilled by inconsistencies" (Jones, 2019). At the same time, the engineering-oriented conception of data has dominated IS (Alaimo et al., 2020), and this may have fallen short of explaining their "intangible nature" (Faulkner and Runde, 2019) in the organizational context. For example, while research identifies an "objective and homogeneous" nature in the big data context of strategizing (Constantiou & Kallinikos, 2015), it also argues that the use context (or "situated practice") is the key variable in this loop, as discussed by Jones (2019) — "unless data are sought, selected, extracted, and interpreted [for a purpose], they cannot inform". Given the diversity and complexity of data use in nature, we recognize the field needs to make further efforts to consolidate and reflect on what has been discussed. Absence of this kind of research could result in "pretheoretical understandings" (Ekbia, 2009) of the entities involved, "with the potential lack of clarity, detail and nuance" (Faulkner and Runde, 2019).

Second, despite the varied ways that data are used to create value in organizations, the approach to conceive and measure data value has not kept pace but remains outdated, and disjointed (Aaltonen et al., 2023; Xu et al., 2023). For years, the data-information-knowledge-action value chain has had a strong presence in IS research. However, the latest developments in data characteristics have transcended this boundary (Abbasi et al, 2016). Grounded from IT/IS value research, data has been consistently regarded as a strategic resource of organizations (Chen et al., 2012; Günther et al., 2017; Surbakti et al., 2020), emphasizing resource allocation and orchestration as the core to realize data value (Ghasemaghaei, et al., 2015; Ghasemaghaei and Calic, 2020; Trieu, 2017). More recent literature also identifies data as an intangible asset contributing to knowledge creation and improving decision-making (Shollo and Galliers, 2016; Côrte-Real et al., 2017). While these perspectives have been instrumental in recognising the impact of data use on business outcomes (Trieu, 2017), it may not suffice to capture the intricate nature of data and their dynamic capabilities in contemporary contexts. In the era of big data, increased volume leads to complexities, not necessarily better decisions (Abbasi et al., 2016); data can now be widely reused and shared without diminishing its value, challenging the fundamental economic assumptions of data value (Jones and Tonetti, 2020). Therefore, scholars argue that any initiatives of value capture and measurement should consider the nature of data use specific to their purposes and contexts (Aral and Weill, 2007; Xu et al., 2023). In a recent discussion in the data research community about "what is missing from research on data in the field of IS" (Aaltonen et al., 2023), scholars advocate that the core of IS research needs to center around exploring the pivotal role of data: "this involves fostering critical and innovative scholarly inquiry into data, spanning the intersections between IS and adjacent fields".

To address these issues in the current body of knowledge, this study conducts a comprehensive literature review aimed at examining the diverse and up-to-date use of data in organizations and integrating a unique set of theoretical assumptions about the characteristics of data. We pose the following research question: *In what ways can data contribute to business objectives, and how can we conceptualize the role of data in organizations?* Our research aims to offer a twofold contribution. First, we broadly examine and synthesize the existing assumptions in the field concerning data. By bridging multiple perspectives and dimensions, we develop a typology that elucidates the versatile roles that data play in organizations, demonstrating the distinctive characteristics associated with each role. Second, we discuss how this elevated treatment of data within the scholarly community can assist practitioners in leveraging this deeper understanding as they seek to effectively monetize, manage, and govern data assets in their organizations.

This article is structured into six core sections. Following this introduction, we explain our perspective in reviewing data in organizations and the stances we adopt in this investigation. Next, we explain the review methodology, and how we proceed to synthesize the versatile roles of data that emerge from the literature. Later, we present a typology of identified roles and characteristics of data, followed by a discussion of practical and theoretical implications of these findings.

A new perspective on data in organizations

Our intention to draw attention to the roles of data has been inspired by Orlikowski and Iacono (2001), who appeal for the need to develop more sophisticated understandings regarding the role of IT artifacts. The recent discussions of Faulkner and Runde (2019) about digital objects have also sparked further reflection. Both emphasize the central role of technology within larger systems in which they operate and interact with. IT artifacts – typically packaged in tangible forms such as hardware and/or software – are always embedded in a context (Benbasat and Zmud, 2003), and are not "natural, neutral, universal, or given" (Orlikowski and Iacono, 2001). According to Faulkner & Runde (2019), digital objects can in some way resemble people in the sense that they hold social positions within larger systems. These positions, among other "nonmaterial qualities", are interconnected, and can significantly shape the social identity of the individuals involved. Hence, it is important to take a holistic view in examining the interplay between key enablers: technology, people, and the organization. The dynamic interactions among them drive outcomes throughout the various stages of use within the organization (Orlikowski and Iacono, 2001; Hevner et al., 2014). We summarize our perspective by highlighting the two critical positions that we take in this investigation.

First, we explore a wide array of data usage in organizations, placing particular emphasis on the purpose and context of use. Robust evidence has shown that data have been used in many ways in organizational contexts and served multiple business objectives (Xu et al., 2023). According to Aral and Weill (2007), a strategic purpose is vital, which highlights what specific business objectives the IT assets are to support within an organization. DeLone and McLean (2003), in their ten-year review of the IS Success Model, contend that "merely stating that more use will yield more value, without considering the nature of this use, is clearly insufficient"; therefore, researchers must also consider "the nature, extent, quality, and appropriateness of the system use". Building upon this perspective, we further unpack our research question into a series of inquiries: *What specific business purposes are served by using data? What business value does data contribute to? In what way is the data value assessed or reflected? What other factors, such as people and organizational practices, may play a role in the value creation?* As we contemplate these questions throughout the literature review, we aim to

comprehend the dynamics at play and the unique value derived from data use.

Furthermore, we examine and reflect on the characteristics of data in organizations, considering any association with their usage. To make sense of these diverse but nuanced phenomena, we have drawn on the theoretical lens of Orlikowski and Iacono (2001)'s framework as a set of "views" — the tool view, the proxy view, the ensemble view, the computational view, and the nominal view — each representing a common set of assumptions about and treatments of information technology in IS research (see Appendix A for details of these views). Over time, scholars have acknowledged the usefulness of these perspectives in conceptualizing technology (Akhlaghpour et al., 2013; Matook and Brown, 2017; Faulkner and Runde, 2019). In addition, data are also relatable to the category of digital artifacts (Kallinikos et al., 2013; Faulkner and Runde, 2019). Digital artifacts, such as documents or media, possess dynamic attributes (Kallinikos et al., 2013). They can have a hybrid of material and non-material components. These qualities create a dependence on the functional relationships between artifacts and their specific contexts and organizations (Faulkner and Runde, 2013, 2019). At different stages of our discussions, we refer to these concepts with an interest in understanding how the specific characteristics of data intersects with broader categories in which data reside. This assists us in reflecting on the characteristics of data with a broad view, rather than confining the discussions narrowly.

Review method and process

In this section, we outline the process and methods used for searching, selecting, and analyzing literature for our review. Our approach to the literature review aligns with the broad-theorizing review guidelines introduced by Leidner (2018). Distinguished by research focus or objective from three other major types— organizing reviews, assessing reviews, and specific-theorizing reviews — the broad-theorizing approach organizes and describes the literature, aiming to use the synthesis and description as a "springboard" for theorizing (Leidner, 2018). By adopting this approach, we assert that our method is integrative in nature, as our efforts are focused on integrating existing knowledge previously dispersed in the literature (Leidner, 2018; Cronin and George, 2023). On the Breslin and Gatrell (2023) "miner-prospector continuum" used to classify the types of literature reviews, our approach can be best described as prospectors, rather than miners. Prospectors may "take a broader view", "venture outside disciplinary boundaries", hoping to introduce new perspectives or conceptualizations. A prospector approach may involve strategies of transferring theories across domains, blending and merging ideas that may seem at first unconnected (Breslin and Gatrell, 2023). We have found these strategies very relatable in our review process.

Given the extensive scope of this study, we strive for comprehensiveness by using a structured process, or using an algorithm, to search and select the literature, considering both conceptual and empirical contributions (Paré et al., 2015; Cronin and George, 2023). Yet, we are also mindful of balancing between a comprehensive set of relevant literature and the quality, importance, and the "level of faith" in reporting them (Cronin and George, 2023). Below, we describe our review process according to the three key steps in which it was conducted.

The first step involves scoping the topic and identifying its boundaries. We began with extensive reading and immersed ourselves in the literature to gain a broad understanding of the focal topic, key concepts, and research streams. To facilitate our scoping process, we created mind maps, which provide a sketch of discussions and key concepts covered in the literature, helping us gauge the breadth of the topic. It was evident that data have been discussed diversely in literature and concepts are interwoven. With a strategic orientation towards leveraging data to improve organizational performance, we focused our study on Information Systems, and neighboring areas such as Business Management, Strategy, Organization, and Economics. The inclusion of publications in these adjacent fields ensures that we did not miss essential concerns and arguments in these cognate disciplines. We also included proceedings of premier conferences in the fields of IS and Management that have occurred within the last five years. This ensures that we do not overlook publications that have yet to be included in journals due to the lengthy review cycle.

We crafted search keywords to cover all related discussions about data use and their associated value or impact for organizations. We used algorithmic search strings that contain variations of keywords: "data business value or strategic value", "data value or valuation", "data (or analytics) and performance (or impact)", "data monetization", "data commodity", and "data marketplace". We compared different sources of literature and made decisions over the databases, and inclusion and exclusion criteria. Some critical considerations of these decisions are explained in the methodology details Appendix (Appendix B, Table B-1).

The second step is literature search and selection. We used Web of Science as the main database, supplemented by AISeL and some journal websites such as MIS Quarterly and The Journal of Strategic Information Systems. The initial search returned 3,508 academic publications with our filtering criteria excluding non-peer-reviewed lecture notes and book chapters. We then conducted a manual screening of each article to identify its relevance to our study, considering several factors. To begin with, we assessed the content by title and abstract, considering its relevance to answering our research questions. While there is a vast related body of literature, our study focuses on investigating the roles and value of data, thereby limiting our search primarily to the data business value domain and the motivational aspect of data use (Surbakti et al., 2020). Specifically, we looked for papers that mentioned a data use purpose or context and possible business outcomes. We then excluded investigations of non-organizational context or overly narrow business scenarios; this helped to restrict the discussions to an organizational setting and enabled mid-range generalizability of our research findings. Given that data have been discussed very broadly, this step excluded many papers from the initial dataset. Furthermore, we conducted a backward and forward snowballing process (Webster and Watson, 2002) to include more literature via a bibliometric co-citation approach. We used seminal papers collected in earlier stages as concept-centric "seeds" to identify those who cited them or those whom they cited, and to identify links in concept development. We also performed targeted searches for missing papers that included more recent adoptions of big data analytics (BDA), AI, and machine learning. The final selection of the sample included 142 papers from 41 peer-reviewed journals and 8 premier conference proceedings. Details about the literature selection process and

composition of the final sample are provided in Appendix B (Table B-1 and Table B-2).

The third step is literature sorting, coding, and analysis. Given our objective to describe a complex data phenomenon and to critically synthesize theoretical arguments, we have come to realize that it is necessary to adopt different analytical methods at different stages of our analysis, situating this vast body of literature. The initial stage is akin to an interpretive qualitative way of organizing literature (Leidner, 2018), aiming to sensitize the phenomena. Specifically, we began by familiarizing ourselves with the setting and making inquiries about themes in a fairly open way: what is going on there? First, we started with a sample of empirical papers and conducted a set of ordered steps of open coding and sorting, with an interest in observing the phenomena of data practices in organizations. We applied a set of criteria in literature coding – such as research methods, keywords, and concepts or theories. We also extracted information and observed aspects of business purposes of data use, people (i.e., users, decision makers, and their roles), and organizational practices (i.e., processes, organizational functions, systems, etc.) in these studies. Through this process, we identified key themes of data use, with the support of representative empirical papers (Paré et al., 2015). We considered empirical studies as our "data", or "proofs" (Alvesson and Kärreman, 2007), that facilitate and motivate critical reflection. The constructed nature of empirical materials enhanced our ability to "challenge, rethink, and illustrate theory" (Alvesson and Kärreman, 2007). Second, we conducted an in-depth content analysis of a selection of conceptual papers in an integrative manner (refer to Appendix C) for papers selected for content analysis), which aimed to explore how the relevant theoretical assumptions can aid in interpretating the phenomena (Paré et al., 2015). During this step, we actively engaged with existing theories, transitioning from inductive to a form of abductive research. This tandem approach to data and existing theories (Alvesson and Kärreman, 2007) helped us integrate concepts and gather evidence to provide a comprehensive treatment of the subject (Schryen et al., 2020; Watson and Webster, 2020). This step allowed us to establish initial categories to formulate our typology by which we seek to define the roles of data in organizations.

Below, we report our key findings from the literature review.

A synthesis — The roles of data in organizations

We observe that data exhibit varied characteristics depending on their usage in different business contexts. To better understand these characteristics, we extensively reference and integrate conceptual discussions across different domains, allowing us to construct a coherent explanation. The theoretical framework by Orlikowski and Iacono (2001), referred to as "the views", offers a valuable perspective for our investigation at a broad level. We also systematically collect and categorize attributes or characteristics of data as discussed in the literature, resulting in a set of meta categories within these overarching viewpoints, which we term "roles of data".

In this section, we unpack our findings by proposing four primary roles that data play in organizations: data as a tool, data as a commodity, data as a practice, and data as algorithmic intelligence.[1] By identifying these roles, we underscore the multifaceted use of data in organizational settings, acknowledging both their traditional roles as well as their evolving and transformative nature in today's data-driven business landscape.

Data as a Tool

The first important role we identify is data as a *tool*. This role is typically recognized when data are used to support decision-making processes in managerial contexts. We construct this meaning of data value from the research theme that addresses the use of data in decision-making.

Theme 1: informing business decision-making

Barkin and Dickson (1977) and Delone and Mclean (1992) were among the pioneers in exploring the use of data within the realm of human decision-making. Drucker (1999) subsequently built upon their work by defining information as "data endowed with relevance and purpose". This perspective finds support among numerous scholars who conceptualize data as unprocessed, raw facts capable of being transformed into accurate and pertinent information (Knox, 2007). Zins (2007) maps the interrelations between data, information, and (human) knowledge, to form a basic block of concepts in the information science field. More recently, Jones (2019) refers to a pyramid, "the Data/Information/Knowledge/Wisdom (DIKW) hierarchy" developed by Rowley (2007) and describes it as "one of the most fundamental, widely recognized and 'taken-for-granted' models in the information and knowledge literatures".

Since 2010 s, many studies in the IS field started to investigate the use of data in organizations and its impact on business results. We present a list of empirical studies as illustrations (Table 1), which reveal that economic and social value can be gained from data through enhanced decision making, and more informed strategizing. Within this context, typical data use scenarios include business planning, product design and ideation, resource allocations, structured management, and operational decisions (Blohm et al., 2013; Brynjolfsson and McElheran, 2016; Shamim et al., 2019). Directly observed benefits of the use relate to the process of knowledge creation, absorbing, and sharing in organizations; thus data value is captured through improved decision-making quality (i.e., customer insights) and more structured management practices (i.e., strategic alignment within organizations) (Brynjolfsson and McElheran, 2016 ; Shollo and Galliers, 2016; Wang and Byrd, 2017). Such data use largely contribute to firm-level outcomes, such as market performance (i.e. sales and market value), customer satisfaction, and engagement (Bekmamedova and Shanks, 2014).

[1] The choice of labelling data as "a tool" recognizes the foundational origins of meaning in alignment with Orlikowski and Iacono's (2001) terminology, and acknowledges the intersection of these roles with information technology in general. By naming the other roles differently – such as data as a "commodity", as a "practice", and as "algorithmic intelligence" – we wish to highlight their distinctions in meaning.

Table 1
Empirical studies about data use for informing decision making (n = 13).

References (in chronological order)	Outlets (in alphabetic order)	Key Words or Concepts	Research Contributions
Blohm, I., Leimeister, JM. and Krcmar, H. (2013) Bekmamedova, N and Shanks, G. (2014) de Vries, A. Chituc, CM. and Pommee, F. (2016) Wang, Y. and Byrd, T.A. (2017) Janssen, M., et al. (2017) Song, PJ., Zheng, CD., Zhang, C. and Yu, XF. (2018) Ghasemaghaei, M. (2019) Shamim, S. et al. (2019) Mantymaki, M, Hyrynsalmi, S. and Koskenvoima, A. (2020) Oesterreich, T. D., Anton, E., & Teuteberg, F. (2022) Roeder, J., Palmer, M. and Muntermann, J. (2022)	Business Information Systems Workshops Decision Support Systems (2) Information & Management (2) * Information Systems Frontiers Journal of Business Research Journal of Enterprise Information Management Journal of Knowledge Management MIS Quarterly Executive (2) *	*Data-driven decision-making, Data (analytics) business value, Business performance, Data-driven competitive advantage, Data insights.*	These papers provide evidence of the business value of data and demonstrate that data play a key role in informing decision-makers. Furthermore, they show that data have an impact on firm performance across different industries and contexts.
Someh, I.A. and Shanks, G.G. (2015) Brynjolfsson, E. and McElheran, K. (2016)	American Economic Review HICSS	*Data-driven decision-making, Firm performance, Business analytics, Data payoffs.*	Empirical evidence shows that data use is associated with improved business results, such as revenue growth, higher productivity, cost savings in operations, and new product ideation and design.

*Denotes the number of publications if more than one.

Improved productivity, market valuation, new product ideation & launches (Blohm et al., 2013) or improved products and services, are also found to be associated with the data use.

The tool view

The tool view acknowledges information technology as an "engineered artifact", which permits the user and the organization to adapt the use of technology according to their specific needs and context (Orlikowski and Iacono, 2001). Just as traditional tools serve as the basis for organized labour, new digital tools have the potential to significantly impact and reshape labour and human activity (Karanasios et al., 2021). These new digital tools can alter, amplify, and transform how work is performed (Faulkner and Runde, 2019; Karanasios et al., 2021).

By considering data as a tool, we direct our attention to its "instrumental effects" (Orlikowski and Iacono, 2001) and its role in "mediation" (Vygotsky, 1999); so that the interaction between individuals (the subject) and their environment (the object) is mediated by tools (Vygotsky, 1999; Miettinen et al., 2009). While using data is more of a human centred activity (Knox, 2007), the user and the organization can vary the needs and purpose of using data, reshape the way of use, and improve the outcome (Orlikowski and Iacono, 2001; Jones, 2019). As a result, the value of data is "subjective" and dependent on the user or designer, determined by their specific needs and "situated practice" (Jones, 2019). Data in this context, similar to other IT artifacts, can be considered separate from the organizational setting; they are "definable", "unchanging", and "independent" (Orlikowski and Iacono, 2001).

Data as a Commodity

The second role we identify is data as a commodity, in which data is traded or sold as goods to generate direct revenue or exchanged for other resources, mostly between organizations. This differentiation sets it apart from other types of data usage that are often limited to within an organization.

Theme 2: data trading

Yuchtman and Seashore (1967) identify that some resources are relatively "liquid" in the traditional economic sense and are readily exchangeable by an organization for resources of other kinds. More recently, Birch et al. (2021) argue that data can be "assetized", and personal data can have measurable and legible value and be potentially converted into future revenue streams. This also applies to situations where data are integrated or bundled with other products or services, thereby contributing to their value-add in the marketplace (Wixom and Farrell, 2019).

A list of empirical studies has demonstrated how data generate revenue through data exchange or direct selling (Table 2). This stream of research represents an emerging but important direction, covering a diverse range of topics, including data liquidity, data commodity, data business models, data marketplaces, and data valuation (Xu et al., 2023).

Under this research theme, data trading emerges as the main scene; the value creation process, mostly inter-organizational, involves a two-sided market coordination between buyers and sellers (Spiekermann, 2019; Parvinen et al., 2020). Investigations also

Table 2
Empirical studies related to data trading (n = 14).

References (in chronological order)	Outlets (in alphabetic order)	Key Words or Concepts	Research Contributions
Sammon, D. and Nagle, T. (2017) Schüritz, R. Seebacher, S. and Dorner, R. (2017) Kühne, B. and Böhmann, T. (2019) Spiekermann, M. (2019) Parvinen, P. et al (2020) Lange, H. E., Drews, P., and Höft, M. (2021)	HICSS Intereconomics ECIS (2) Communications of the Association for Information Systems ICIS	*Data-driven business models, Data monetization, Value proposition, Revenue models, Data value map*	These studies investigate how companies can capture value from data-driven services. They provide broad-level guidance on the design and selection of appropriate revenue models and, thus, inspire new forms of revenue generation from the use of data.
Najjar, MS. and Kettinger, WJ. (2013) Smith, G., Ofe, H.A. and Sandberg, J. (2016) Alfaro, E., et al. (2019) Aaltonen, A., Alaimo, C. and Kallinikos, J. (2021)	MIS Quarterly Executive (2) HICSS Journal of Management Information Systems	*Data monetization, Data value, Data commodities, Digital innovation, Data management*	These case studies provide empirical evidence of how organizations generate revenue from their data assets by selling and trading. They also provide insights for organizational practices to commodify or monetize data.
Chen, Y.J. and Huang, K.W. (2016) Birch, K., Cochrane, D. T. and Ward, C. (2021) Mehta, S., et al. (2021) Ye, H., et al. (2021)	Big Data & Society Information Systems Research (2) Journal of Management Information Systems	*Data assets, Data marketplace, Data pricing, Valuation, Personal data, Digital content, Pricing plans, Revenue streams.*	These studies discuss data pricing and valuation methods, as well as recommendations for accounting and regulations in selling personal data or digital content.

include data-driven business models or service offerings (Alfaro et al., 2019; Najjar and Kettinger, 2013; Schüritz et al., 2017; Lange et al., 2021; Ye et al., 2021) as well as data pricing strategies or policies (Chen and Huang, 2016; Mehta et al., 2021). A typical value creation pathway may include data generation, preparation or productization, packaging (i.e., through modularization or decontextualization), brokering or tendering, valuation, transmission, and consumption. Data can be transmitted through a system "bearer" – hardware or software, or through APIs, and/or IT infrastructure (i.e., federated machine learning). The interaction and coordination are deemed important between the supply side and the demand side. The aim of such interactions is to make successful transactions between the sellers and the buyers. For example, a pricing mechanism of the data marketplace is described as follows:

> *"Firms such as Data & Sons, Dawex, and Lo-tame cater to both sides of the market: sellers list their data sets on the platform and buyers purchase data from one or more sellers. Here, the pricing of the data can be done by the seller or by the platform on behalf of the seller. Other firms such as BIG and Narrative enable auction-based purchase and sale of data —buyers browse data sets from different sellers and bid their valuations, and sellers manage their transactions by selecting buyers of their choice."* (Mehta et al., 2021)

Recent studies indicate a potentially greater bargaining power for buyers (Agarwal et al., 2019; Ray et al., 2020). Nevertheless, considering the economic principle of supply and demand, we anticipate that bargaining power in transactions can fluctuate. A more advanced data marketplace or brokerage system would contribute to balancing interactions within the two-sided market and promoting transparency and fairness in the pricing of data goods (Ray et al., 2020; Agarwal et al., 2019).

The proxy view

The proxy view recognizes the adoption of "surrogate measures" to represent the essential attributes of objects, or as "symbols" (Orlikowski and Iacono, 2001). In the context of data being bought or sold, the common characteristic is the attribution of a monetary value during transactions.

Viewing data as a commodity, we focus on the "tradable" nature of data in several ways. First, data can be considered liquid assets as they can be easily converted into revenue at a low cost (Birch et al., 2021; Wixom et al., 2021; Piccoli et al., 2022). Certain types of data may exhibit a higher level of liquidity compared to others (Piccoli et al., 2022). Second, data is recognized as comprising distinct atomic components or data tokens that lend themselves to easy repurposing, recombination, and restructuring for the creation of new solutions (Kallinikos et al., 2013; Alfaro et al., 2019; Aaltonen et al., 2021). As such, research suggests that data possess characteristics as being editable, interactive, reprogrammable, distributable, and exchangeable (Aaltonen et al., 2021; Alaimo and Kallinikos, 2022). Third, when data are bought or sold, they are regarded separable from organizations. They can be decoupled from their hardware or software infrastructure, due to the separability nature of "bitstrings", which is a type of digital object as defined by Faulkner and Runde (2019).

While discussing trading data in a marketplace, research takes into account some of the key characteristics of data goods: data are "freely replicable, and combinatorial (i.e., features have overlapping information)" (Agarwal et al., 2019). The concept of modularity,

a design feature that denotes the extent to which a system's components can be detached and reassembled (Schilling, 2000), has been examined in relation to the reuse and recombination of datasets (Piccoli et al., 2022). Scholars believe that data products with such features likely lead to larger gains by reusing and repurposing (Wixom and Ross, 2017; Piccoli et al., 2022).

With the unique nature of tradability, data product offerings can now be available "off the shelf" through direct sales or via the platforms of a data marketplace (Spiekermann, 2019; Parvinen et al., 2020). Furthermore, data can be used by multiple firms simultaneously without reducing their value. This characteristic of data being "nonrival" means that its use by one firm does not diminish its usability for others; and the nonrivalry of data leads to increasing returns, as highlighted by Jones and Tonetti (2020). While traditional IT artifacts can also be measured by their monetary value, typically represented in terms of the cost invested in them (Orlikowski and Iacono, 2001), data has clearly surpassed this notion by generating direct revenue as a profit driver (Birch et al., 2021). Hence, research proposes that data should be documented in a company's accounting balance sheet as corporate assets to precisely reflect the value of data in business management (Birch et al., 2021). However, research also highlights that this practice has not yet been embraced, despite the success of many Big Tech companies that have capitalized on their user data (Birch et al., 2021). This remains a critical issue to address in the space of data reuse by digital platforms (Gregory et al., 2021).

Data as a Practice

The third role we identify is data as a practice, which represents repeatable patterns of activities that leverage data for continuous improvements in business operations, and for optimizations of products or services (Wade and Hulland, 2004; Grover et al, 2018).

Theme 3: enabling innovation and learning

Our collection of empirical studies comprises a substantial proportion of research focused on the role of data in digital innovation and learning, which is an influential yet continuously evolving area of study. Numerous real-world case studies highlight the significance of data in driving digital innovations and facilitating organizational learning. Noteworthy examples include Netflix (Gomez-Uribe and Hunt, 2015), Westpac (Anand, Sharma and Coltman, 2016), Microsoft (Wixom and Farrell, 2019), and Uber (Farronato et al., 2020).

Table 3
Empirical studies related to data use for innovation and learning (n = 17).

References (in chronological order)	Outlets (in alphabetic order)	Key Words or Concepts	Research Contributions
Chen, DQ., Preston, DS. and Swink, M. (2015) Gupta, M. and George, JF. (2016) Côrte-Real, N., Oliveira, T., and Ruivo, P. (2017) Krishnamoorthi, S. and Mathew, SK. (2018) Muller, O., Fay, M. and vom Brocke, J. (2018) Olszak, CM. and Zurada, J. (2020)	Information & Management (2) Information Systems Management Journal of Business Research Journal of Management Information Systems (2)	*Big data analytics* *Analytical capabilities* *Dynamic capabilities* *Organizational value creation* *Competitive advantage*	These articles provide insights into how organizations leverage business analytics to build unique competitive advantages and improve organizational performance. Each article focuses on a specific aspect or industry context, contributing to the broader understanding of the business value derived from analytics capabilities.
Zolnowski, A., Christiansen, T., & Gudat, J. (2016) Chen, H.M., Schutz, R., Kazman, R. and Matthes, F. (2017) Mikalef, P; Boura, M; Lekakos, G; Krogstie, J. (2019) Ghasemaghaei, M. and Calic, G. (2020) Dong, JQ. and Yang, CH. (2020) Duan, YQ., Cao, GM. and Edwards, JS. (2020) Saleem, H.,et al. (2021)	ECIS British Journal of Management European Journal of Operational Research Information & Management Journal of Business Research Journal of Enterprise Information Management MIS Quarterly Executive	*Data-driven innovation* *Value co-creation* *Business process improvement* *Business model innovation*	These studies revolve around the importance of data-driven innovations, business model transformation, value realization from digital data streams, deepening the understanding of how organizations leverage big data in improving innovation performance.
Gomez-Uribe, C.A. and Hunt, N. (2015) Shollo, A. and Galliers, R.D. (2016) Fink, L. Yogev, N. and Even, A. (2017) Ferraris, A., Mazzoleni, A., Devalle, A. and Couturier, J. (2019)	ACM Transactions on Management Information Systems Information & Management Information Systems Journal Management Decision	*Data-driven value creation* *Business intelligence & analytics (BI&A)* *Organizational learning* *Organizational knowing* *Knowledge management*	These studies specifically investigate how data-enabled learning contributes to organizational learning and knowledge management.

As depicted in Table 3, we have identified three clusters of studies that examine various aspects of the phenomena (despite the fact that the concepts in these studies are commonly interwoven) – such as data-driven organizations, and data-driven innovation or learning. The initial cluster of studies takes a broad view and explores the impact of business analytics or BDA on organizational competitiveness (Chen et al., 2015; Krishnamoorthi and Mathew, 2018; Olszak and Zurada, 2020). This line of research highlights the unique set of capabilities or competence that data contribute to, which forms competitive advantages of an organization (Xu et al., 2023). The second cluster of studies focuses specifically on digital innovations, or data-driven innovation, and the role data play during the process. These investigations provide insights of how organizations leverage big data to enhance their innovation performance (Duan et al., 2020; Mikalef et al., 2019; Zolnowski, 2016). Lastly, the third cluster of studies centres on *data-enabled learning*, the role of big data in facilitating learning processes within organizations. These studies offer deeper insights into how data can be leveraged to improve organizational knowledge (Shollo and Galliers, 2016; Fink and Even, 2017; Ferraris et al, 2019).

Overall, these three clusters of research studies collectively enhance our understanding of how organizations harness data to drive learning and innovation in business management and improve the competitive advantage of an organization. When data are utilized for innovation and learning, particularly in areas like product design, service improvement, and process management, it typically follows a structured business process or workflow. This process involves ordered steps from data generation to knowledge creation (Shollo and Galliers, 2016; Grover et al., 2018), or improving their services by leveraging their *data network effect* in a platform (Hagiu & Wright, 2020). By adhering to this practice, organizations can sustain their competitiveness in products and services (Wade and Hulland, 2004; Grover et al., 2018).

The synergy between BDA and various other resources and capabilities within an organization introduces a novel strategy for maintaining a competitive edge (Aral and Weill, 2007; Grover 2018). Specifically, how data is instrumented and used, as well as the support provided by infrastructure and management systems, significantly impact results. In the context of organizational learning, Shollo and Galliers (2016) emphasize the significance of two key abilities: the capacity to initiate problem articulation and constructive discussions, as well as the skill to curate data effectively. As such, the "three gears" – data, insights, and actions – work together to generate strategic business value through BDA development and implementation (Grover et al., 2018).

The ensemble view

The ensemble view emphasizes the interactions and relationships between IT artifacts and the groups involved in their construction, implementation, and use (Orlikowski and Iacono, 2001). Scholars emphasize that learning and innovation are social and cultural phenomena in nature, not just individual mental processes (Gherardi, 2000). Learning through participation, or learning by doing, highlights that learning occurs within everyday practices, even without our conscious awareness. Gherardi (2000) argues that knowledge is "neither in the head nor as a commodity", but "being in the world" whereby subjects (people) and objects (knowledge) are "indistinguishable", existing within ongoing action and interaction.

In the context of digital innovation and learning, we consider data as an organizational practice, by identifying their distinctive characteristics in three aspects. First, data represents complex interactions between people and technology, emphasizing "doing" rather than "being" (Orlikowski and Iacono, 2001). This is because, from a practice perspective, "knowledge [generation] is dynamic, provisional, and continually reproduced and negotiated through participation"; it is mediated by data, and rooted in contextual interactions (Gherradi, 2000). However, it is worth noting that data-enabled learning now surpasses the traditional learning-by-doing notion in which humans have more control. In large digital platforms, the *data network effect* emerges and drives outcomes through

Table 4
Empirical studies related to data use for modelling and automation (n = 12).

References (in chronological order)	Outlets (in alphabetic order)	Key Words or Concepts	Research Contributions
Davenport, T.H. and Ronanki, R. (2018) May, A., et al. (2020) Rinta-Kahila, T. et al. (2021) Fügener, A., Grahl, J., Gupta, A., and Ketter, W. (2021) Lou, B. and Wu, L. (2021) Van den Broek, E., Sergeeva, A. and Huysman, M., (2021) Marjanovic, O., Cecez-Kecmanovic, D. and Vidgen, R. (2021) Mikalef, P. and Gupta, M. (2021) Shollo, A., Hopf, K., Thiess, T. and Müller, O. (2022) Fanti, L., Guarascio, D. and Moggi, M. (2022) Beese, J. and Fahse, T. (2023) Sagodi, A., Dremel, C. and Van Giffen, B. (2023)	*ECIS (2) ** *European Journal of Information Systems* *Harvard Business Review* *ICIS* *Information & Management* *Journal of Information Technology* *Journal of Industrial and Business Economics* *Journal of Strategic Information Systems* *MIS Quarterly (3)*	*Artificial intelligence, Algorithmic decision-making, Machine learning, Digital innovation, Process automation, AI value creation.*	These studies explore the role of AI for process improvement, digital innovation, as well as the value creation process and mechanisms. Studies identify that data in different forms of AI and ML applications create value and impact business results by automating business processes and algorithmic decision-making.

data sharing, as theorized by Hagiu and Wright (2020). Examples are given such as Google Maps and similar data-driven platforms, where organizations can learn from their user data to improve product offerings or services, therefore the value of their service or product is leveraged by the amount of data they own or operate on.

Second, data is organizationally embedded, with various elements such as processes and routines, management support, and corporate culture, all working together to leverage data for competitiveness (Grover et al., 2018). The process of data value creation also requires an organizational capacity to support. Being organizationally embedded implies that data operate within a complex and dynamic social context, intricately connected to the conditions of their use (Orlikowski and Iacono, 2001; Grover et al., 2018). It entails recurring sets of activities or procedures that function both as means to accomplish organizational tasks and as mechanisms for storing and accessing knowledge about the most effective ways to achieve those objectives (Aral & Weill, 2007). Data in this role exhibit characteristics similar to IT artifacts in terms of being adaptable, integrable, and complex (Matook and Brown, 2017), and there is a "coevolutionary adaptation" cycle through "learning by doing" (Grover et al., 2018). Studies indicate that rendering data in digital platforms transforms our way of generating knowledge (Alaimo and Kallinikos, 2022), or creating user value (Hagiu and Wright, 2020).

Third, data possess a social identity within the communities in which they originate. Faulkner & Runde (2019) define this social identity in digital objects with "social positions" that locate them as components in larger systems. We argue that data as a practice are not isolated entities either, but rather integral components within a broader social framework. They may occupy specific positions and relationships within the digital ecosystem, and these positions are vital to understanding the social identities of the individuals or organizations associated with them.

Data as Algorithmic Intelligence

Lastly, our final collection of studies (Table 4) describes the role of data in supporting process automation and algorithmic decision-making, thereby generating value for organizations at an unprecedented pace and with remarkable versatility, potentially surpassing our imaginations. This is achieved through various forms of aritificial intelligence (hereafter referred to as "AI") and machine learning applications in business.

Theme 4: automating and algorithmic modelling

The first serious thinking about AI is often traced back to the Dartmouth summer workshop in 1956, where McCarthy and colleagues defined the project as making a machine behave intelligently like a human (Berente et al., 2021). They believed that all aspects of intelligence, including learning, could be "formalized and reproduced in a machine". This notion has later been widely cited and claimed (Dick, 2019). Recent definitions of AI emphasize its ability to learn (Castelvecchi, 2016), or its capability of mimicking human capabilities and skills (Brynjolfsson and Mitchell, 2017). Today the vision of the field of AI goes beyond acting as a decision support system for human work. Instead, it advances to create artificial agents to replace human labour (Schuetz and Venkatesh, 2020). AI can act on behalf of humans by employing natural language processing chatbots, intelligent agents, and machine learning to interact with employees and customers (Davenport et al., 2018). A recent example is ChatGPT, which utilizes deep learning algorithms to generate human-like responses to natural language inputs (Radford et al., 2019).

Observed as an emergent topic, recent research has identified that AI presents unprecedented opportunities for organizations in improving productivity, creating intelligent products, and enhancing services. The typical use of AI in organizations involves two main types: process automation and algorithmic decision-making (ADM). Process automation is the most common business need fulfilled by AI, i.e., utilizing robotic process automation to automate administrative and financial tasks, both in digital and physical realms (Davenport et al., 2018). Moreover, algorithms play a vital role in processing rules and making decisions without human intervention. Powerful algorithms are built and deployed to work autonomously on behalf of humans and make decisions (Newell and Marabelli, 2015). In certain cases, automated decision-making algorithms can override human decisions (Enholm et al., 2021; Galliers et al., 2017; Shollo et al., 2022). However, such usage can lead to unintended consequences, including discrimination and social inequality, which can have harmful effects on society (Rinta-Kahila et al., 2021). To address these concerns, data curation and human oversight are essential to ensure the effectiveness and appropriateness of AI-driven outcomes (Sadiq et al., 2022).

The computational view

The computational view recognizes the ability of IT artifacts in representing, manipulating, storing, retrieving, and transmitting information (Orlikowski and Iacono, 2001). Technology can be represented here through algorithmic endeavors to build new or enhance existing computational systems that can support some human activity (Orlikowski & Iacono, 2001). Earlier, Zuboff (1985) proposed two faces of "intelligent" technology — "automate" and "informate"—and their respective implications for organizations. IT can be applied to automate operations, substituting human labor and expertise with technology. This allows the replication of human labour at a reduced cost and with more control and continuity. Technology can also simultaneously generate information about the automated tasks, a process referred to as "informate".

In a recent MIS Quarterly editorial article on the adoptions of AI in organizations (Berente et al., 2021), the editors argue that AI is not simply a technology or a defined set of technologies. Instead, it represents "an ever-evolving frontier of computational advancements that references human intelligence in addressing ever more complex decision-making problems". Noting there remains a lack of precision in the conceptualization of AI, "the openness and uncertainty surrounding its meaning and usage actually provides greater flexibility for theoretical exploration of AI and its management" (Berente et al., 2021).

When considering data as algorithmic intelligence, we associate the role with the computational view of IT artifacts (Orlikowski

and Iacono, 2001) in a general sense. More importantly, we recognize several important characteristics that have set data apart in this role.

The first characteristic is the constantly evolving capability of data in automation and augmentation. AI exhibits an automated, self-trained, and deeper learning capacity in machines (Davenport et al., 2018; Berente et al., 2021; Baird and Maruping, 2021). AI systems go beyond mimicking humans and autonomously undertake tasks according to designed rules. Rather, AI serves to augment and enhance human abilities, leading to synergistic collaboration between humans and machines (Baird and Maruping, 2021). Though some may argue automation and augmentation emerge at different stages of use, many scholars believe they are intertwined in processing information and reaching an optimal decision (Teodorescu et al., 2021; Thi Ngoc and Elbanna, 2023).

The second characteristic is a new way that machines interact with humans, and a new way of delegation. The primacy of human agency in the human-technology interactions has been shifting over time (Karanasios et al., 2021), and AI has a greater autonomy than any other intelligent IT artifact or "agentic IS artifacts" (Baird and Maruping 2021; Berente et al., 2021). Now, autonomous agents undertake tasks in ways that are not necessarily delegated by humans; instead, they often rely on their information processing abilities to autonomously select optimal solutions (Berente et al., 2021). As operating with algorithmic decision-making models, machines are no longer passive tools waiting to be used and are no longer always subordinate to the human agent; they can now assume responsibility for tasks with ambiguous requirements and for seeking optimal outcomes under uncertainty (Baird and Maruping, 2021). Their autonomy is increasingly generative (Seidel and Berente, 2020). On the one hand, people delegate to autonomous agents in various ways (Baird and Maruping, 2021). On the other hand, autonomous agents also direct and control humans (Kellogg et al., 2020; Möhlmann et al., 2021). Murray et al. (2021) refer to this in terms of different types of "conjoined agency," which involves different types of configurations depending on whether the human or the algorithm takes the control of selecting the actions or developing the

Table 5
A typology: four major roles that data play in organizations.

	Data as a Tool	Data as a Commodity	Data as a Practice	Data as Algorithmic Intelligence
Conceptualization	Data are engineered artifacts to serve intended business purposes. The user and the organization can vary the needs and purposes of using data, reshape the ways of use, and improve performance.	Data are represented by their monetary value, and their common feature of tradability in the marketplace.	Data in this role represent an ensemble, which emphasizes the interactions, and relationships, between IT artifacts and the groups involved during use.	This role of data focuses on the ability of algorithmic computation that continuously evolves and simulates human intelligence.
Purpose or Context of Use	***Informing*** decision making	***Data Trading***	***Enabling*** innovation and learning	***Automating*** and algorithmic modelling
Value Creation Pathways or Mechanisms	Data contribute to enhancing the effectiveness of decision-making and elevating individuals' cognitive levels through processes of informing, sharing knowledge, and absorption.	Data generate revenue through methods such as direct selling, exchange for other resources, or service bundling, often within an inter-organizational context.	Data contribute to organizational capabilities by "participating" in a designed business process, leading to continuous improvements of service or product offerings.	Data exhibit their computational capabilities through information processing, task automation, and decision modelling, ultimately contributing to the reduction or enhancement of human labour.
The Interplay (between people, technology, and organization)	A human centred activity. The user is a key enabler in the value creation process. Data can be separable from an organization.	More inter-organizational, the value creation process involves a coordination of the two-sided market, i.e., buyers, and sellers. Data can be decontextualized and separable from organizations.	People constitute just one component of this ensemble; more significantly, it is the repetitive practice, encompassing a repeatable pattern of activities, that integrates data into ongoing business processes. This integration ensures the enduring sustainability of organizations' competitive advantages.	A machine centred activity. With greater autonomy, the machine can even act without direct human intervention.
Characteristics or Attributes	Emphasis on the instrumental effects. Independent of social setting. Relative to user and the context of use. Situated, contextualized. Human instrumentation driven.	Can be represented by a monetary value. A set of distinct atomic components that can be "repurposed, recombined and restructured." Editable, interactive, reprogrammable, distributable, and exchangeable. Can be decoupled from hardware or software infrastructure.	Organizationally embedded and context situated. Data-enabled learning. Data network effect. Imitable. Integrable, complex, and adaptable. Have a social identity.	Mimicking human capabilities and skills. Can be automated, augmented, and self-trained. Deeper learning capacity. Inscrutable. Can be reused and recombined. Can form a complex network of technologies, each with its own design objectives.

rules.

The third characteristic is the potential in connectivity and scalability. AI technology provides connectivity and possibilities for exchange among users. Through automation or augmentation (Raisch and Krakowski, 2021), AI enables substantial scaling of data-enabled learning from the digital imprints of users when they interact with devices, facilities, people, and organizations in their everyday life (Gregory et al., 2021). The data network effect (Gregory et al., 2021) therefore constitutes the foundation for the "platform AI capability", enabling AI systems to learn from a broader spectrum of information, thereby enhancing insights and decision-making prowess. The modular architecture of data networks, fostering a complex amalgamation of technologies, each with its distinct design objectives (Gregory et al., 2021), renders it even more adaptable and integratable than before. This enhanced adaptability amplifies functionality and facilitates reuse and recombination in diverse contexts (Gregory et al., 2021; Raisch and Krakowski, 2021; Piccoli et al., 2022).

Lastly, it is important to note that AI is more inscrutable than any other "intelligent" IT artifact as emphasized by scholars (Baird and Maruping, 2021; Berente et al., 2021). This inscrutability brings both unknown benefits and risks. The knowledge generated by AI systems is not solely produced by humans but also by machines, and there is a possibility that certain aspects may become incomprehensible to humans. This raises concerns about ethical harms that may arise from the use of AI. As organizations leverage the expanding capabilities of AI, it becomes crucial for us to establish mechanisms to control and mitigate any potential negative consequences and ensure responsible and accountable use of technology.

These characteristics highlight the role of data as algorithmic intelligence with its generative and collaborative nature, a shifting agency between humans and machines, its scalability potential, and inscrutability. With the rapidly advanced technologies, data as algorithmic intelligence essentially help automate and informate human activity and knowledge in many new ways (Berente et al., 2021).

A typology and its application

Typology Description

This section provides an integrated view towards the roles of data in organizations. Table 5 presents detailed descriptions of these roles with a typology, which is achieved through extracting a common set of attributes across the categories, and further abstracted to provide a higher-order theoretical structure (Paré et al., 2015). This typology helps capture the multifaceted data practices in organizations and identify key aspects to observe or manage. Through our synthesis, several recurring first-order constructs emerge as the defining dimensions to differentiate these types:

First, *the purpose of data use* is observed as a determinant factor to differentiate the roles. We delineate the diverse business objectives that data serve in an organization and encapsulate them using concise keywords such as *informing, trading, enabling,* and *automating*. These distinct purposes represent the key themes, as set out above, that data are used for in organizations; they influence the resultant business impact or value, as well as the way they are constructed, instrumented, or used. *Second*, *value creation pathways or mechanisms* can also set these roles apart. While many data-use cases in the literature identify a common set of business outcomes such as improved market value and productivity, which may lack a link to the use contexts, our focus centres on a more directly observable array of benefits arising from specific usage. This includes enhancing the effectiveness of decision-making, generating revenue, optimizing processes, and automating. These directly observable benefits are essential to understand value creation pathways or mechanisms with more nuanced, disaggregated contexts. To some extent, they may help frame the process of creating internal and competitive value, which is regarded as unexplained in the IS value research (Schryen, 2013). *Third*, *the interplay* between people, technology, and organization, can diverge across each role. For instance, data serve as a tool to inform business decisions, underscoring the human-centric nature of the decision-making process. Here, the value of data predominantly hinges on the end-users. In contrast, data can assume the form of algorithmic intelligence, where algorithms play a substantial role in generating value, and the interaction between individuals and technology becomes more centred around machines rather than humans. Similarly, data can be likened to a commodity, signifying their potential to be separated from an organization, decontextualized, and then recontextualized. Conversely, when data is viewed as a practice, it becomes deeply embedded within the organizational fabric, rendering it less separable from the organizational context. *Fourth*, *the characteristics of data* exhibit distinct facets when examined through the lens of different roles. These characteristics may be drawn from broader categories, such as IT artifacts or digital artifacts, or they may possess unique attributes and qualities stemming from the specific data use cases. These distinguishing traits are instrumental in our classification of these roles.

In light of these observations, we have drawn together and synthesized concepts and evidence from different parts of the literature to create a more coherent framework (Cornelissen, 2017) – a typology to define four major roles of data (Table 5). Through the development of this typology, the effort is aimed to illustrate "how phenomena can hang together" (Sandberg and Alvesson, 2021). According to Cornelissen (2017), a typology typically interrelates different dimensions to flesh out a novel explanation with new constructs or casual relationships which pushes our thinking about a subject.

Applying the Typology

Next, we apply this typology in a reflective literature mapping analysis, which covers all papers that we have initially sourced from the systematic search and selection stage. We focus on 142 papers published from year 2010 to the present, a period during which data applications have become more dynamic and varied. We assign role labels to the samples, — "tool", "commodity", "practice", and "AI" (in an abbreviated form) —with the aim of observing how this typology may be used to classify the literature and what, if any, patterns

can be observed over time. In general, a strong alignment is observed between the conceptualizations presented in the typology and the existing literature. A significant majority, exceeding 90% of the articles, can be categorized and labelled without difficulty, indicating a high degree of correspondence; Table 6 presents a summary of the literature mapping result, with details in Appendix D. However, it is worth noting that certain articles may address multiple roles of data, highlighting the complexity and interconnectedness of these roles within the research landscape. To simplify the analysis, we present the primary role of data in each study. For example, in the study by Chen et al. (2012), which discusses various developments of data usage, we categorize it as discussing the primary role of data as a tool. Similarly, Constantiou and Kallinikos (2015) primarily focus on the tool perspective, Grover et al. (2018) emphasizes the practice role, and Shollo et al. (2022) delve into AI. We made these judgments by matching their key arguments with definitions in our typology but also by counting the frequency of keyword mentions. For example, we looked for mentions of "decision-making" to identify the tool role and mentions of "innovation" to highlight the practice role, and vice versa. The coding and mapping were independently carried out by two researchers until a consensus was reached.

During the observed timeline from 2010 to the present, the literature mapping identifies that the roles of data in organizations demonstrate evolving trends. Topics regarding the tool role of data remained relatively stable with minor fluctuations during the years. In contrast, the practice perspective witnessed a significant rise starting from 2015, reaching its peak in 2022. The emergence of the commodity role has been more pronounced in the recent years, especially after 2015. AI-related discussions displayed a consistent upward trend, particularly after 2020. In general, there was an overall increase in the number of studies conducted, reflecting a growing recognition of the importance of data and their diverse applications within organizational contexts.

Discussion

Implications for Research

Given that we have extensively integrated concepts and evidence across domains, here we explain how our findings engage with some scholarly discussions.

First, the concept of "intelligent" technology has primarily revolved around the capabilities to "automate" and "informate" for a few decades since it was first proposed by Zuboff (1985). By identifying novel yet significant roles that data play, we propose an expansion of this notion. This entails their capacities for "informing," "trading," "enabling," and "automating", both within and across organizations. This expansion of capacities has been supported by robust evidence that we have drawn from the literature. Moreover, some characteristics distinctly set data apart from other IT artifacts, not only by their usage, effect, and character, but also in the way data interact with people and organization. Notably, some emergent roles of data, i.e., data as a commodity or as algorithmic intelligence exhibit a reduced dependence on human orchestration in the value creation trajectory and develop an increased objectivity in their quality during transitions. This increased objectivity allows for greater possibility to decouple data from the organizational context, which, in turn, enhances the adaptability of data for reuse and repurposing (Günther et al., 2022).

Second, our research introduces fresh perspectives to the central tenets of IT/IS value research, specifically in how we define and perceive the value of data. We contribute to the literature by demonstrating that the purpose of data use determines the role data play in organizations and the value they create. While the conventional understanding in the field has largely centered on data as a resource, which provides an overarching explanation of their impact on business outcomes, our findings add more nuanced, fine-grained contexts in which data value is generated and captured. "The many faces of data", as we have depicted, provides explanations for the dynamic, diverse, and complex nature of data value. Through this research, we identify that it is imperative to avoid generalizing or directly transferring assumptions about data from the IT/IS value research. Our research underpins the scholarly assertions that the value of data should not be perceived as fixed or unchanging akin to other IT assets (Aaltonen et al., 2023). Instead, the actual usage of

Table 6
Number of publications describing the (primary) role of data in organizations.

	Number of Publications Describing the (Primary) Role of Data in Organizations					
Year of Publication	Tool	Commodity	Practice	AI	Not Applicable	Total
2010					1	1
2011	1					1
2012	1					1
2013	1	1				2
2014	2					2
2015	1		2	1	2	6
2016	2	2	4			8
2017	4	2	4	2	1	13
2018	1	1	6	2		10
2019	3	4	2			9
2020	1	3	8	1	2	15
2021	2	6	3	13	2	26
2022	5	1	12	8	1	27
2023	2	3	4	6	2	17
2024			2	2		4
Grand Total	26	23	47	35	11	142

data (or IT) pays off and drives business impact (Devaraj and Kohli, 2003). While more evidence shows that data can also be used and reused by many organizations simultaneously, which leads to increasing returns (Jones and Tonetti, 2020), there are potentially large gains out of data being used broadly.

Implications for Practice

Defining the versatile roles data play in organizations and highlighting their distinctions holds significant meaning for practice. Each of the four roles have important implications. Below, we highlight a few of these implications for organizations. In doing so, we intend to shine a light on how the distinctions in each role can reshape the ways how organizations utilize data and assess their value. Then we offer our perspective what, collectively, these roles mean for organizations in the way they manage data and why an overarching data strategy is important.

Through leveraging different roles of data, organizations have the potential to reshape and expand the avenues through which they utilize data for diverse business objectives. These benefits include enhancing managerial decision-making, generating direct revenue by trading data, driving innovation, and automating business processes. It is important to recognize that these roles vary not only in the way they create value but also in the nature of use. Therefore, each role may demand a different set of supportive organizations capabilities. *The tool role* of data emphasizes the individuals using them, so organizations must focus on developing data related skillsets and competencies (Grover et al., 2018): for example, improving data literacy of the workforce to remove barriers in using data. *The practice role* of data emphasizes the importance of embedding data within a repeatable process and formalizing it. People are just one part of this ensemble; more importantly, it is the repeatable practice that embeds data within continuous business processes, ensuring sustainable competitive advantages for organizations (Grover et al., 2018). Useful practices include building centralized data platforms to democratize enterprise data to a broader audience, enabling greater access by a wide range of users, both experts and non-experts (Labadie et al., 2020; Lefebvre et al., 2021). These practices help reduce potential barriers in individuals' competency or restrictions in data access.

In addition, attention should also be drawn to the emergent roles of data that arise from their diverse and continuously evolving usages in contemporary organizations. *The commodity role* of data is characterized by its tradable nature, indicating that data should no longer been seen as a cost for organizations but a source of profit. In fact, data can be considered one of the most liquid assets that an organization possesses (Piccoli et al., 2022). Therefore, commodifying an organization's data assets has emerged as a strategically promising business model. Nowadays, data marketplaces have become readily available, and buying and selling processes have been extensively discussed, making data trading more practical than ever before. Data as *algorithmic intelligence* opens an unimaginable potential of data in replacing human labour and enhancing our ability in solving complex problems. To effectively leverage value from AI, organizations should focus on developing AI-related capabilities. This involves integrating AI initiatives with existing organizational capabilities – such as talent, management support, and process management – to create new and complementary human–machine capabilities (Krakowski et al., 2023; Shollo et al., 2022). At the same time, due to the nature of inscrutability in AI applications, organizations must address the risks associated with AI by developing control systems to mitigate potential harms and ensure ethical and responsible AI deployment (Berente et al., 2021; Rinta-Kahila et al., 2021).

Collectively, while grasping the unparalleled opportunities that data offer, it also introduces new challenges, urging organizations to adapt their strategies in data monetization, management, and governance. First, since each of these roles differs significantly in terms of its use context and the value creation pathways, we contend that a generalized approach to data monetization will not be effective. Instead, organizations should develop a comprehensive data strategy that addresses diverse sources of data value and the unique pathways to unlock the specific value associated with each role. This involves creating tailored business models and effectively linking, selecting, and allocating organizational resources to support the strategic purposes of monetizing data.

Second, recognizing the nuanced characteristics and contexts of data within organizations is instrumental in driving the reconfiguration and transformation of their dynamic capabilities. Data-driven capabilities have been widely recognized as a primary source of competitive advantage and have demonstrated the potential to yield superior performance. The development of organizational capabilities requires effective resource allocation, which entails both formal and informal processes (Dosi, 2000). Organizations also need to upskill their workforce in data literacy and provide supporting organizational mechanisms to effectively support the strategic objective of building data-related capabilities (Aral and Weill, 2007; Grover et al., 2018).

Third, developing effective data governance strategies is equally vital for organizations to address potential risks and ethical concerns. These strategies must strike a balance between safeguarding data privacy and security while facilitating the process of capturing, using, and reusing data within and across the organization(s). On one hand, establishing clear processes and policies that cover the entire data lifecycle, from creation or acquisition to storage or consumption, is crucial. This comprehensive approach enhances information resilience and enables organizations to promote data consistency, eliminate data silos, and facilitate efficient data-driven value creation in business management (Surbakti et al., 2020; Sadiq et al., 2022). On the other hand, data privacy and security demand attention, particularly in the context of data reuse via sharing and selling (Surbakti et al., 2020). While social license provides a guardrail when the human involvement is low (Sadiq et al., 2022), regulatory considerations are equally important to legitimize the use of data in AI initiatives (Gregory et al., 2021).

In brief, organizations require an integrated data strategy that not only harnesses the potential of data but also sustains data-related capabilities and safeguards legitimate data use. This comprehensive approach ensures that data value is fully utilized, and organizations are prepared for long-term success while maintaining ethical and responsible data use.

Limitations and Future Research Opportunities

We acknowledge that there are limitations to our research. First, our typology hinges on the intangible nature of data within the business contexts and organizational systems within which data are operated. This means we pay less attention to ontological questions, such as their technical properties as an "entity"; epistemological questions, such as the approach of acquisition of scientific knowledge; or semantic questions, such as the meaning of word "data" in different contexts and situations (Gregor, 2006; Sandberg and Alvesson, 2021). As discussed by Aaltonen et al. (2023), these perspectives pose multi-faceted complexity for both research and practice to address the role of data in organizations. IS research needs to "be open to accommodating different perspectives in conceptualizing data as no single definition will be able to account for the complex entanglements of data with economic and social life". Second, because our investigation focuses on the social context and interactions in organizations, we have not covered enough about the engineering-oriented conception of data. This may limit the applicability of our findings in the real-world business management context in which we acknowledge technical aspects of data are equally important to consider.

Last, while we have focused on the strategic value of data in organizations and have demonstrated their positive impact on organizational outcomes (as suggested by most of the literature), we also recognize that data have other aspects. Earlier, Feldman and March (1981) argue that organizations tend to systematically gather more information than they use, driven by the symbolic significance of data rather than its instrumental value. It is not uncommon to see some organizations gather large amounts of data and generate reports because it is expected or because it projects an image of "intelligence" (Feldman and March, 1981), even if the data is rarely used to inform decisions. Recently, Essén et al. (2022) explore how business users ignore data that they themselves produced. This behavior underscores the ritualistic and symbolic use of data, which often results in data being treated as a byproduct or even a waste of organizational resources. While this cluster of interesting discussions can be potentially addressed by the nominal view of Orlikowski and Iacono's (2001) framework, we suggest a deeper investigation is needed to explore these "dark sides" (Rana et al., 2022) of data and their implications for organizational practices.

Briefly, we discuss opportunities for future research. The typology of data we derive in this paper will open a new area of research in the field of Information Systems. First, by understanding the "many faces of data" in organizations, the value of data should not be measured and generalized in a blanket manner, rather it should be contextualized based on specific use purposes. By gaining a more nuanced understanding of data use, researchers can advance the theoretical connection between data use and value. This will enable us to determine the most effective methods for utilizing and managing data in contemporary organizations. Second, the four types of data roles we conclude in this paper do not seem to have clear-cut boundaries in the way they are performed in practice; the context of use, and the value they represent, can be interwoven in many organizational practices. For example, when data are used as a tool to support business decision-making, a repeatable process of data-driven decision making can contribute to building a data-related organizational capability that is sustainable. Likewise, a repeatable process of preparing data, trading data as commodity, or a recuring practice utilizing data in algorithmic decision-making, not only requires an organizational capacity across many business functions to support, but also contributes to forming a unique set of organizational capabilities. There appears to be an interchanging value flowing between the four data roles, which needs to be further addressed by future research.

In conclusion, we have undertaken a comprehensive synthesis to organize existing literature and provide an integrated view of the roles data play in organizations. Our investigation centres on the past two decades, a period characterized by the most dynamic evolutions in which organizations have gone through different stages of data adoption to support their business goals, ranging from "static" data to BDA and AI. We present a synthesis of the characteristics of data that correspond to data usage in organizations. By bridging these dimensions, we propose a typology to conceptualize four major roles of data, which also holds practical relevance for application.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This paper stems from the first author's PhD dissertation. The authors thank her committee: Andrew Burton-Jones, Ping Wang, Gianluca Dermatini, and Saeed Akhlaghpour. The authors thank Mats Alvesson, and Wai Fong Boh for their developmental feedback. Additionally, the authors have benefited from the review process and presentations at ECIS 2023 (Kristiansand), AoM 2024 (Chicago), the Doctoral Consortium at ACIS 2023 (Wellington), the Doctoral Consortium of the CTO Division at AoM 2024 (Chicago). The authors also received constructive comments from several paper development workshops at The University of Queensland, organized by Jorgen Sandberg, Paula Jarzabkowski, and Stan Karanasios, which were instrumental in refining this work.

The authors thank Yolande E. Chan (EIC), Shan L Pan (SE), Jyoti Choudrie (AE), and the three anonymous referees for their invaluable guidance during the revision process at The Journal of Strategic Information Systems.

This research was supported (partially or fully) by the Australian Government through the Australian Research Council's Industrial Transformation Training Centre for Information Resilience (CIRES) project number IC200100022.

Appendix

Appendix A:. Conceptualization of IT artifacts by Orlikowski and Iacono, 2001

.

	Excerpts of Argument
Tool View	Technology is the engineered artifact, expected to do what its designers intend it to do. As such, what the technology is and how it works are seen to be largely technical matters – matters that are separate, definable, and unchanging, matters over which humans have control.
Proxy View	Described in terms of one or more (usually quantitative) surrogate measure taken to represent the essential feature(s).
Ensemble View	Emphasis on the interactions, and relationships, between IT artifacts and the groups involved in their construction, implementation, and use. Enmeshed with the conditions of its use—hence, our label "embedded system".
Computational View	Emphasis on the computational powers of IT artifacts; their ability to represent, manipulate, store, retrieve, and transmit information.
Nominal View	IT artifacts mentioned in name only or not at all.

Appendix B:. Methodological details of literature Search, selection and analysis

.

Table B1
Review Process, Tasks and Sub-tasks, Results or Outputs.

Task	Sub-Task	Result / Output
STEP 1. SCOPE THE REVIEW		
1. 1 Sensitize the topic	We engaged in extensive reading to immerse ourselves in the literature and gain a broad picture of the focal topic. This stage has helped us make important methodological decisions in sourcing and selecting the literature.	Mind maps of the research topic. A collection of seminal papers.
1.2 Identify relevant fields of research	We narrowed down the fields of study to focus on the most pertinent literature. The field of business information systems hosts a vast body of knowledge on this topic. Some scholars join the conversation from management perspectives; thus, the literature can span several fields, including Business Management, Strategy, and Organization. Notably, Economics scholars also contribute an important stream of discussions, such as data trading and pricing.	Information Systems, Business Management (and Strategy), Organization, and Economics.
1.3. Determine sources	We compared different sources of literature, i.e., coverage, quality, sub-categories, and timeliness of inclusion etc. We determined to focus on Web of Science as our main database, supplemented by AISeL and some journal websites.	Web of Science (the main database) AISel Online Library Journal Websites (e.g., JSIS)
1.4 Decide search keywords	Keyword search queries are crafted and used in the online searches. As we refined the search queries, we had some important considerations. First, "data" and "value" is largely correlated with neighboring fields such as computer science engineering. To focus on business outcomes rather than technical, we used data "business value" or "strategic value" to differentiate the terms and narrow down searches. Second, we considered all possible variations of some keywords, such as monetization (or monetisation) and adopted coding techniques to reflect these variations in search strings such as "moneti?ation". We made rounds of refinements in these search techniques and carefully compared the search results, with internal peer reviews during the process.	**Keywords (**with codes to reflect possible variations**):** data business value or strategic value, data moneti?ation, data value or data valuation, data use & performance, data marketplace, data commodit*, analytics value or impact (Please note search strings are varied by each source depending on the set up of the search engines).
STEP 2. CONDUCT THE LITERATURE SAERCH		
2.1 Initial hit	a) Web of Science (on topic) Category: (Computer Science) Information Systems; Management Filter out: lecture notes, book chapters, book reviews, call for papers. b) AIS eLib (on abstract, title, and subject) Removed duplicates.	n = 3,508
2.2 Journal title filter	We filtered by journal title to focus on publications of Information Systems, and premier journals of business management, Organization, and Economics (not in specific area, such as business finance, marketing, or HR).	n = 1,227
2.3 Manual screening of papers	We did a hand pick based on title, abstract, plus additional texts if necessary to select paper that are relevant to answer our research	n = 109

(*continued on next page*)

Table B1 (*continued*)

Task	Sub-Task	Result / Output
	questions. Inclusion criteria: clear and central focus on data usage in and between organizations, covering all types of data applications (ie. BA&BI, AI and machine learning). Exclusion criteria: a) general discussions of technology use and value discussions (e.g., IT or IS business value); b) operational aspects of data management in organizations such as data governance, data quality curation etc.)	
2.4 Backward and forward search	Additional picks by backward and forward search through tracing citation references.	n = 33
2.5 Finalize data set	Decide a final sample and re-check the selections.	n = 142
STEP 3. ANALYZE THE LITERURE AND PRESENT RESULTS		
3.1 Literature sorting, coding, and grouping	Empirical papers are analyzed inductively through literature sorting, coding, and grouping. This process has enabled us to observe the key themes related to data usage in practice.	Findings are presented in section A Synthesis – The Roles of Data in Organizations
3.2 Content analysis	Conceptual papers are analyzed using an integrative approach through content analysis, which aims to gather evidence supporting key assumptions derived from the literature.	

Table B2
Literature Selection – by Source Title.

	SOURCE TITLE	NO. OF PAPERS
PEER- REVIEWED JOURNALS (111 PAPERS)	INFORMATION & MANAGEMENT	15
	INFORMATION SYSTEMS FRONTIERS	10
	MIS QUARTERLY	7
	JOURNAL OF STRATEGIC INFORMATION SYSTEMS	7
	JOURNAL OF MANAGEMENT INFORMATION SYSTEMS	6
	DECISION SUPPORT SYSTEMS	6
	MIS QUARTERLY EXECUTIVE	5
	JOURNAL OF INFORMATION TECHNOLOGY	4
	EUROPEAN JOURNAL OF INFORMATION SYSTEMS	4
	ACADEMY OF MANAGEMENT REVIEW	4
	INFORMATION SYSTEMS MANAGEMENT	3
	MANAGEMENT DECISION	3
	JOURNAL OF BUSINESS RESEARCH	3
	JOURNAL OF ENTERPRISE INFORMATION MANAGEMENT	3
	INFORMATION SYSTEMS RESEARCH	3
	JOURNAL OF STRATEGY AND MANAGEMENT	2
	BRITISH JOURNAL OF MANAGEMENT	2
	BIG DATA AND COGNITIVE COMPUTING	1
	INFORMATION AND ORGANIZATION	1
	EUROPEAN JOURNAL OF OPERATIONAL RESEARCH	1
	INFORMATION SYSTEMS JOURNAL	1
	INDUSTRIAL MARKETING MANAGEMENT	1
	SCIENCE INSIGHTS	1
	NATIONAL BUREAU OF ECONOMIC RESEARCH	1
	STRATEGIC MANAGEMENT JOURNAL	1
	JOURNAL OF MANAGEMENT ANALYTICS	1
	COMMUNICATIONS OF THE ASSOCIATION FOR INFORMATION SYSTEMS	1
	HARVARD BUSINESS REVIEW	1
	COMMUNICATIONS OF THE ASSOCIATION FOR INFORMATION SYSTEMS	1
	JOURNAL OF THE KNOWLEDGE ECONOMY	1
	JOURNAL OF COMPUTER INFORMATION SYSTEMS	1
	BIG DATA & SOCIETY	1
	ACM TRANSACTIONS ON MANAGEMENT INFORMATION SYSTEMS	1
	MIT SLOAN MANAGEMENT REVIEW.	1
	JOURNAL OF INDUSTRIAL AND BUSINESS ECONOMICS	1
	RAND JOURNAL OF ECONOMICS	1
	AMERICAN ECONOMIC REVIEW	1
	JOURNAL OF KNOWLEDGE MANAGEMENT	1
	ORGANIZATION SCIENCE	1
	INFORMATION TECHNOLOGY & MANAGEMENT	1
	INTERECONOMICS	1
CONFERENCE PROCEEDINGS (31 PAPERS)	ECIS	10
	HICSS	6
	ICIS	5
	ACIS	4

(*continued on next page*)

Table B2 (*continued*)

SOURCE TITLE	NO. OF PAPERS
PACIS	2
BUSINESS INFORMATION SYSTEMS WORKSHOPS	2
ACM EC 2019	1
AMCIS	1

Appendix C:. Conceptual papers used for the content analysis (by year Order)

.

Author(s), Year	Title	Outlet	Key Contributions / Justifications for Inclusion
Chen, H. et al. (2012)	Business intelligence and analytics: from big data to big impact	MIS Quarterly	This introduction to the MIS Quarterly Special Issue on Business Intelligence Research provides a framework that identifies the evolution of business intelligence and analytics, applications, and emerging research areas. BI&A. BI&A 1.0, BI&A 2.0, and BI&A 3.0 are defined and described in terms of their key characteristics and capabilities.
Constantiou, D. and Kallinikos, J. (2015)	New games, new rules: big data and the changing context of strategy	Journal of Information Technology	This research contends that the dynamics involved in the generation and distribution of large volumes of data lead to substantial changes in how organizations create and utilize pertinent information. This transformative process introduces noteworthy consequences for the realm of strategic decision-making, along with the way we manage data and information within the organizational strategy framework.
Newell, S. and Marabelli, M. (2015)	Strategic opportunities (and challenges) of algorithmic decision-making: A call for action on the long-term societal effects of "datification"	Journal of Strategic Information Systems	The paper delves into the realm of algorithmic decision-making driven by extensive data analysis, particularly data stemming from digitized devices. The utilization of such data enhances an organization's capacity for refined decision-making and discrimination, offering potential benefits but also raising societal implications.
Galliers, R. D. et al. (2017)	Datification and its human, organizational and societal effects: The strategic opportunities and challenges of algorithmic decision-making	Journal of Strategic Information Systems	This commentary on the JSIS Special Issue on datification focuses on two key themes selected from among the many topics discussed by Special Issue authors: 1) the debate over algorithmic intelligence versus human intelligence, and 2) the consequences of strategic performance systems.
Günther et al. (2017)	Debating big data: A literature review on realizing value from big data	Journal of Strategic Information Systems	This study conducts an in-depth systematic review of IS literature on the topic and identify six debates central to how organizations realize value from big data and identify two socio-technical features of big data that influence value realization: portability and interconnectivity.
Trieu, VH. (2017)	Getting value from Business Intelligence systems: A review and research agenda	Decision Support Systems	This study reviews a collection of empirical Information System (IS) studies about the processes of organizations obtaining business value from BI systems.
Brynjolfsson, E., & Mitchell, T. (2017)	What can machine learning do? Workforce implications	Science Insights	The recent advances in machine learning (ML) accelerate the pace of automation itself and mimics human capabilities and skills. This paper discusses key implications for the workforce, what the current generation of ML systems can and cannot do.
Kitchens, B., Dobolyi, D., Li, JJ. and Abbasi, A. (2018)	Advanced Customer Analytics: Strategic Value Through Integration of Relationship-Oriented Big Data	Journal of Management Information Systems	This study investigates organizational agility in combining rich data across the organization to deploy analytics that sense and respond to customers in a dynamic environment. A key challenge in achieving this agility lies in the identification, collection, and integration of data across functional silos both within and outside the organization.

(*continued on next page*)

(*continued*)

Author(s), Year	Title	Outlet	Key Contributions / Justifications for Inclusion
Grover, V., Chiang, R. H., Liang, T. P., & Zhang, D. (2018)	Creating strategic business value from big data analytics: a research framework	Journal of Management Information Systems	This study offers a framing of BDA value by extending existing frameworks of information technology value, then illustrate the framework through BDA applications in practice. The framework is then discussed in terms of its ability to study constructs and relationships that focus on BDA value creation and realization.
Sun, S., Cegielski, C.G., Jia, L. and Hall, D.J. (2018)	Understanding the Factors Affecting the Organizational Adoption of Big Data	Journal of Computer Information Systems	This study identifies the main factors affecting the organizational adoption of big data, based on the retrieval and review of relevant papers in the business intelligence & analytics (BI&A) literature published during the period 2009–2015.
Agarwal, A., Dahleh, M. and Sarkar, T. (2019)	A marketplace for data: An algorithmic solution	ACM EC 2019	The authors propose a mathematical model for a two-sided data market by addressing the challenge stemming from the very nature of data as an asset: (i) it is freely replicable; (ii) its value is inherently combinatorial due to correlation with signal in other data; (iii) prediction tasks and the value of accuracy vary widely; (iv) usefulness of training data is difficult to verify a priori without first applying it to a prediction task.
Jones (2019)	What we talk about when we talk about (big) data	Journal of Strategic Information Systems	The author presents a critical analysis of a number of widely held assumptions about data in general and big data in particular. The distinction between "data in principle" and the "data in practice" is discussed, questioning their foundational objectivity.
Faroukhi, A.Z., et al. (2020)	An adaptable big data value chain framework for end-to-end big data monetization	Big Data and Cognitive Computing	This paper presents a comprehensive review of existing Big Data Value Chain models relying on some definitions and theoretical foundations of data monetization and offers a data monetization model to support many co-creation contexts along the BDVC.
Ray, J., Menon, S; and Mookerjee, V. (2020)	Bargaining over Data: When Does Making the Buyer More Informed Help?	Information Systems Research	Investigating data marketplace negotiation, this study explores the role of demonstrations in mitigating uncertainty and bias, utilizing a generalization of Nash bargaining.
Wiener, M., Saunders, C. and Marabelli, M. (2020)	Big-data business models: A critical literature review and multiperspective research framework	Journal of Information Technology	This review paper assesses the state of the big-data business model literature and identify shortcomings in this literature. The authors derive and discuss five central research perspectives (supply chain, stakeholder, ethics, national, and process), providing guidance for future research and theory development in the area.
Surbakti, F. et al. (2020)	Factors influencing effective use of big data: A research framework	Information & Management	This paper conducts a systematic literature review to understand and consolidate from literature the potential factors that can influence the effective use of big data.
Hagiu, A. and Wright, J. (2020)	Data-enabled learning, network effects and competitive advantage	RAND Journal of Economics	The authors model competition between firms which improve their products through learning from customer data. The model allows us to analyze factors affecting an incumbent's competitive advantage such as the shape of firms' learning functions and the extent of data accumulation. The study also explores the implications of public policies towards data sharing, user privacy and data acquisitions.
Enholm, I.M., et al. (2021)	Artificial Intelligence and Business Value: a Literature Review	Information Systems Frontiers	This study provides a systematic literature review to explain how organizations can leverage AI technologies in their operations and elucidate the value-generating mechanisms. The analysis synthesizes the current literature and highlights: (1) the key enablers and inhibitors of AI adoption and use; (2) the typologies of AI use in the organizational setting; and (3) the first- and second-order effects of AI.
Klee, S., Janson, A., Leimeister, JM. (2021)	How Data Analytics Competencies Can Foster Business Value- A Systematic Review and Way Forward	Information Systems Management	This study systematically reviews the literature aiming to support organizations by showing relevant competencies and achievable business values.

(*continued on next page*)

(*continued*)

Author(s), Year	Title	Outlet	Key Contributions / Justifications for Inclusion
Gregory, R.W., Henfridsson, O., Kaganer, E. and Kyriakou, H. (2021)	The role of artificial intelligence and data network effects for creating user value	Academy of Management Review	The authors theorize about a new category of network effects—data network effects—that has emerged from advances in artificial intelligence and the growing availability of data. A platform exhibits data network effects if, the more that the platform learns from the data it collects on users, the more valuable the platform becomes to each user.
Fast, V., Schnurr, D., & Wohlfarth, M. (2021)	Regulation of data-driven market power in the digital economy: Business value creation and competitive advantages from big data	Journal of Information Technology	The study reviews the academic literature and show empirical evidence for business value creation from big (user) data. The authors characterize three regulatory approaches to govern market power and competition in data-driven digital markets: (i) empowering consumers, (ii) data openness, and (iii) limiting data scale.
Raisch, S., & Krakowski, S. (2021)	Artificial Intelligence and Management: The Automation-Augmentation Paradox	Academy of Management Review	The authors explore the automation and augmentation concepts in the management domain. Whereas automation implies that machines take over a human task, augmentation means that humans collaborate closely with machines to perform a task. Taking a normative stance, scholars advise organizations to prioritize augmentation, which they relate to superior performance.
Berente, N., Gu, B., Recker, J., & Santhanam, R. (2021)	Managing artificial intelligence	MIS Quarterly	This editorial review introduces AI as an ever-evolving frontier of computational advancements in addressing complex decision-making problems. The authors argue about three interdependent facets of AI—autonomy, learning, and inscrutability—in the ongoing quest to push the frontiers of performance and scope of AI.
Monino, J. L. (2021)	Data Value, Big Data Analytics, and Decision-Making	Journal of the Knowledge Economy	This paper proposes a conceptual framework that information is considered an interpreted flow of data, while knowledge is created and organized from this flow of information.
Teodorescu et al. (2021)	Failures of Fairness in Automation Require a Deeper Understanding of Human-ML Augmentation	MIS Quarterly	The authors introduce a typology of augmentation for fairness consisting of four quadrants: reactive oversight, proactive oversight, informed reliance, and supervised reliance. They identify intersections with previous IS research and distinct managerial approaches to fairness for each quadrant.
Sturm, T., et al (2021)	Coordinating Human and Machine Learning for Effective Organizational Learning	MIS Quarterly	This study delves into how organizations can effectively synchronize human learning with machine learning (ML) for comprehensive learning outcomes. Drawing upon a series of simulations using agent-based models, the researchers ascertain that ML has the potential to alleviate the need for intensive human exploratory learning, which aims to uncover novel ideas within the organization.
Jamwal, A., et al (2022)	Machine learning applications for sustainable manufacturing: a bibliometric-based review for future research	Journal of Enterprise Information Management	Through a systematic and bibliometric-based review in the applications of machine learning (ML) techniques for sustainable manufacturing (SM), this paper highlights how manufacturing industries can benefit from ML techniques when applied to address SM issues.
Alaimo, C., & Kallinikos, J. (2022)	Organizations decentered: data objects, technology and knowledge	Organization Science	Data's pervasive nature transforms it from an administrative element to a resource through which organizations understand and act upon challenges, making data objects integral to organizational knowledge. Users, customers, products, and physical machines rendered as data objects become the technical and cognitive means through which organizational knowledge, patterns, and practices develop.
Krakowski, S., Luger, J., and Raisch, S. (2022)	Artificial Intelligence and the Changing Sources of Competitive Advantage.	Strategic Management Journal	Using chess as a controlled setting for studying competitive interactions, this paper identifies that AI adoption triggers interrelated substitution and complementation dynamics, which make humans' traditional competitive capabilities obsolete, while creating new sources of persistent heterogeneity when humans interact with chess engines.

(*continued on next page*)

(*continued*)

Author(s), Year	Title	Outlet	Key Contributions / Justifications for Inclusion
Aaltonen, A., Alaimo, C., Parmiggiani, E., Stelmaszak, M., Jarvenpaa, S. (2023)	What is Missing from Research on Data in Information Systems? Insights from the Inaugural Workshop on Data Research.	Communications of the Association for Information Systems	This article summarizes discussions from a workshop where senior scholars identified three main research themes: expanding traditional data conceptualizations, examining the relationship between data and meaning, and exploring new data management and governance approaches. The article also links these themes to future research directions.
Zhang, X., et al (2023)	How to monetize data: An economic analysis of data monetization strategies under competition	Decision Support Systems	This study develops a game- theoretic model to investigate the impact of direct data monetization on competition between firms. The results show that in equilibrium, high-value firms with competitive advantages in product or service value often sell less data than low-value firms, implying that high-value firms rely less on direct data monetization than low-value firms.

Appendix D:. Applying the typology — Literature mapping results (by year Order)

.

DATA AS A TOOL (n=26 PAPERS)
Shanks, G., & Sharma, R. (2011). Creating value from business analytics systems: the impact of strategy. PACIS.
Chen, H., et al. (2012). Business intelligence and analytics: from big data to big impact. MIS Quarterly.
Blohm, I., Leimeister, J. M., & Krcmar, H. (2013). Crowdsourcing: How to Benefit from (Too) Many Great Ideas. MIS Quarterly Executive.
Bekmamedova, N., & Shanks, G. (2014). Social Media Analytics and Business Value: A Theoretical Framework and Case Study. HICSS.
Sharma, R., Mithas, S., & Kankanhalli, A. (2014). Transforming decision-making processes: A research agenda for understanding the impact of business analytics on organizations. European Journal of Information Systems.
Constantinou, I. D., & Kallinikos, J. (2015). New games, new rules: big data and the changing context of strategy. Journal of Information Technology.
Brynjolfsson, E., & McElheran, K. (2016). The rapid adoption of data-driven decision-making. American Economic Review.
de Vries, A., Chituc, C. M., & Pommee, F. (2016). Towards Identifying the Business Value of Big Data in a Digital Business Ecosystem: A Case Study from the Financial Services Industry. Business Information Systems Workshops.
Wang, Y., & Byrd, T. A. (2017). Business analytics-enabled decision-making effectiveness through knowledge absorptive capacity in health care. Journal of Knowledge Management.
Janssen, M., et al. (2017). Factors influencing big data decision-making quality. Journal of Business Research.
Trieu, V. H. (2017). Getting value from Business Intelligence systems: A review and research agenda. Decision Support Systems.
Pappas, I. O., et al. (2017). Social Media and Analytics for Competitive Performance: A Conceptual Research Framework. Business Information Systems Workshops.
Song, P. J., Zheng, C. D., Zhang, C., & Yu, X. F. (2018). Data analytics and firm performance: An empirical study in an online B2C platform. Information & Management.
Ghasemaghaei, M. (2019). Does data analytics use improve firm decision-making quality? The role of knowledge sharing and data analytics competency. Decision Support Systems.
Shamim, S., et al. (2019). Role of big data management in enhancing big data decision-making capability and quality among Chinese firms: A dynamic capabilities view. Information & Management.
Jones (2019). What we talk about when we talk about (big) data. Journal of Strategic Information Systems.
Mantymaki, M., Hyrynsalmi, S., and Koskenvoima, A. (2020). How Do Small and Medium-Sized Game Companies Use Analytics? An Attention-Based View of Game Analytics. Information Systems Frontiers.
Delgosha, M. S., Hajiheydari, N., and Fahimi, S. M. (2021). Elucidation of big data analytics in banking: a four-stage Delphi study. Journal of Enterprise Information Management.
Monino (2021). Data value, big data analytics, and decision-making. Journal of the Knowledge Economy.
Roeder, J., Palmer, M. and Muntermann, J. (2022). Data-driven decision-making in credit risk management: The information value of analyst reports. Decision Support Systems.

(*continued on next page*)

(*continued*)

DATA AS A TOOL (n=26 PAPERS)
Oesterreich, T.D., Anton, E., Teuteberg, F. (2022). What translates big data into business value? A meta-analysis of the impacts of business analytics on firm performance. Information & Management.
Peterson, J; et al. (2022). How managers approach data analytics: A typology through a resource orchestration perspective. Management Decision.
Chen, L.F. et al (2022). IT-business alignment, big data analytics capability, and strategic decision-making: Moderating roles of event criticality and disruption of COVID-19. Decision Support Systems.
McCartney, S. and Fu, N. (2022). Bridging the gap: Why, how, and when HR analytics can impact organizational performance. Management Decision.
Dennehy, D., Conboy, K. and Babu, J. (2023). Adopting learning analytics to inform postgraduate curriculum design: Recommendations and research agenda. Information Systems Frontiers, 25(4), pp.1315-1331.
Bagheri, S. and Zwering, M. (2023). Business Intelligence and analytics success factors: A case study in the financial sector. HICSS.
DATA AS A COMMODITY (n=23 PAPERS)
Najjar, M. S., & Kettinger, W. J. (2013). Data Monetization: Lessons from a Retailer's Journey. MIS Quarterly Executive.
Smith, G., Ofe, H. A., & Sandberg, J. (2016). Digital Service Innovation from Open Data: Exploring the Value Proposition of an Open Data Marketplace. HICSS.
Chen, Y. J., & Huang, K. W. (2016). Pricing data services: Pricing by minutes, by gigs, or by megabytes per second? Information Systems Research.
Schüritz, R., Seebacher, S., & Dorner, R. (2017). Capturing value from data: revenue models. HICSS.
Sammon, D., & Nagle, T. (2017). The Data Value Map: A framework for developing shared understanding on data initiatives. ECIS.
Kühne, B., and Böhmann, T. (2018). Representing Data-Driven Business Models - Towards Extending the Business Model Canvas. ACIS.
Alfaro, E., Bressan, M., Girardin, F., Murillo, J., Someh, I., and Wixom, B. H. (2019). BBVA's Data Monetization Journey. MIS Quarterly Executive.
Spiekermann, M. (2019). Data Marketplaces: Trends and Monetisation of Data Goods. Intereconomics.
Kühne, B., and Böhmann, T. (2019). Data-driven business models - building the bridge between data and value. ECIS.
Agarwal, A., Dahleh, M., and Sarkar, T. (2019). A marketplace for data: An algorithmic solution. ACM EC 2019.
Parvinen, P., et al. (2020). Advancing Data Monetization and the Creation of Data-based Business Models. Communications of the Association for Information Systems.
Faroukhi, A. Z., et al. (2020). An adaptable big data value chain framework for end-to-end big data monetization. Big Data and Cognitive Computing.
Ray, J., Menon, S., and Mookerjee, V. (2020). Bargaining over Data: When Does Making the Buyer More Informed Help? Information Systems Research.
Birch, K., Cochrane, D. T., and Ward, C. (2021). Data as an asset? The measurement, governance, and valuation of digital personal data by Big Tech. Big Data & Society.
Mehta, S., Dawande, M., Janakiraman, G., and Mookerjee, V. (2021). How to Sell a Data Set? Pricing Policies for Data Monetization. Information Systems Research.
Ye, H., Yang, X., Wang, X., and Stratopoulos, T. C. (2021). Monetization Of Digital Content: Drivers Of Revenue On Q&A Platforms. Journal of Management Information Systems.
Lange, H. E., Drews, P., and Höft, M. (2021). Realization of Data-Driven Business Models in Incumbent Companies: An Exploratory Study Based on the Resource-Based View. ICIS.
Aaltonen, A., Alaimo, C., and Kallinikos, J. (2021). The Making of Data Commodities: Data Analytics as an Embedded
Wixom, B.H., Piccoli, G. and Rodriguez, J. (2021). Fast-track data monetization with strategic data assets. MIT Sloan Management Review.
Zhang, X., Yue, W. T., Yu, Y., & Zhang, X. (2023). How to monetize data: An economic analysis of data monetization strategies under competition. Decision Support Systems.
Lee, D., Nam, K., Han, I. and Cho, K., 2022. From free to fee: Monetizing digital content through expected utility-based recommender systems. Information & Management, 59(6), p.103681.
Ofe, H., De Reuver, M., Nederstigt, B. and Janssen, M., 2023. Data analytics platforms: Value propositions and adoption challenges for small hospitality businesses. HICSS.
Wang, F., Zhang, Z., Ma, X., Zhang, Y., Li, X. and Zhang, X., 2023. Paths to open government data reuse: A three-dimensional framework of information need, data and government preparation. Information & Management, 60(8), p.103879.
DATA AS A PRACTICE (n=47 PAPERS)
Chen, D. Q., Preston, D. S., & Swink, M. (2015). How the Use of Big Data Analytics Affects Value Creation in Supply Chain Management. Journal of Management Information Systems.
Gomez-Uribe, C. A., & Hunt, N. (2015). The Netflix Recommender System: Algorithms, Business Value, and Innovation. ACM Transactions on Management Information Systems.
Zolnowski, A., Christiansen, T., & Gudat, J. (2016). Business model transformation patterns of data-driven innovations. ECIS.
Anand, A., Sharma, R., & Coltman, T. (2016). Four Steps to Realizing Business Value from Digital Data Streams. MIS Quarterly Executive.

(*continued on next page*)

(*continued*)

DATA AS A TOOL (n=26 PAPERS)
Gupta, M., & George, J. F. (2016). Toward the development of a big data analytics capability. Information & Management.
Shollo, A., & Galliers, R. D. (2016). Towards an understanding of the role of business intelligence systems in organizational knowing. Information Systems Journal.
Côrte-Real, N., Oliveira, T., & Ruivo, P. (2017). Assessing business value of Big Data Analytics in European firms. Journal of Business Research.
Fink, L., Yogev, N., & Even, A. (2017). Business intelligence and organizational learning: An empirical investigation of value creation processes. Information & Management.
Breidbach, C. F. (2017). How do Fintech Service Platforms Facilitate Value Co-Creation? An Analysis of Twitter Data. ICIS.
Chen, H. M., Schutz, R., Kazman, R., & Matthes, F. (2017). How Lufthansa Capitalized on Big Data for Business Model Renovation. MIS Quarterly Executive.
Krishnamoorthi, S., & Mathew, S. K. (2018). Business analytics and business value: A comparative case study. Information & Management.
Muller, O., Fay, M., & vom Brocke, J. (2018). The Effect of Big Data and Analytics on Firm Performance: An Econometric Analysis Considering Industry Characteristics. Journal of Management Information Systems.
Popovic, A,. Hackney, R,. Tassabehji, R. and Castelli, M. (2018). The impact of big data analytics on firms' high value business performance. Information Systems Frontiers.
Kitchens, B., Dobolyi, D., Li, J. J., & Abbasi, A. (2018). Advanced Customer Analytics: Strategic Value Through Integration of Relationship-Oriented Big Data. Journal of Management Information Systems.
Grover, V., Chiang, R. H. L., Liang, T. P., & Zhang, D. S. (2018). Creating Strategic Business Value from Big Data Analytics: A Research Framework. Journal of Management Information Systems.
Sun, S., Cegielski, C. G., Jia, L., and Hall, D. J. (2018). Understanding the Factors Affecting the Organizational Adoption of Big Data. Journal of Computer Information Systems.
Mikalef, P., Boura, M., Lekakos, G., and Krogstie, J. (2019). Big Data Analytics Capabilities and Innovation: The Mediating Role of Dynamic Capabilities and Moderating Effect of the Environment. British Journal of Management.
Ferraris, A., Mazzoleni, A., Devalle, A., and Couturier, J. (2019). Big data analytics capabilities and knowledge management: impact on firm performance. Management Decision.
Gupta, S., et al. (2020). Achieving superior organizational performance via big data predictive analytics: A dynamic capability view. Industrial Marketing Management.
Ghasemaghaei, M., and Calic, G. (2020). Assessing the impact of big data on firm innovation performance: Big data is not always better data. Journal of Business Research.
Olszak, C. M., and Zurada, J. (2020). Big Data in Capturing Business Value. Information Systems Management.
Dong, J. Q., and Yang, C. H. (2020). Business value of big data analytics: A systems-theoretic approach and empirical test. Information & Management.
Farouk, F. M., and Siew, E. G. (2020). Exploring the Role of Big Data Analytics in Driving Financial Innovation: An Affordance Actualisation Perspective. ACIS.
Duan, Y. Q., Cao, G. M., and Edwards, J. S. (2020). Understanding the impact of business analytics on innovation. European Journal of Operational Research.
Wiener, M., Saunders, C., and Marabelli, M. (2020). Big-data business models: A critical literature review and multiperspective research framework. Journal of Information Technology.
Hagiu, A., and Wright, J. (2020). Data-enabled learning, network effects, and competitive advantage. RAND Journal of Economics.
Saleem, H., et al. (2021). Big data use and its outcomes in the supply chain context: the roles of information sharing and technological innovation. Journal of Enterprise Information Management.
Fast, V., Schnurr, D., & Wohlfarth, M. (2021). Regulation of data-driven market power in the digital economy: Business value creation and competitive advantages from big data. Journal of Information Technology.
Lefebvre, H., Legner, C., & Fadler, M. (2021). Data democratization: Toward a deeper understanding. ICIS.
Alaimo, C., & Kallinikos, J. (2022). Organizations decentered: Data objects, technology, and knowledge. Organization Science.
Gao, J. and Sarwar, Z. (2022). How do firms create business value and dynamic capabilities by leveraging big data analytics management capability?. Information Technology and Management, pp.1-22.
Arias-Pérez, J., Coronado-Medina, A. and Perdomo-Charry, G. (2022). Big data analytics capability as a mediator in the impact of open innovation on firm performance. Journal of Strategy and Management, 15(1), pp.1-15.
Awan, U., Bhatti, S.H., Shamim, S., Khan, Z., Akhtar, P. and Balta, M.E. (2022). The role of big data analytics in manufacturing agility and performance: moderation–mediation analysis of organizational creativity and of the involvement of customers as data analysts. British Journal of Management, 33(3), pp.1200-1220.
Cetindamar, D., Shdifat, B. and Erfani, E. (2022). Understanding big data analytics capability and sustainable supply chains. Information Systems Management, 39(1), pp.19-33.

(*continued on next page*)

(*continued*)

DATA AS A TOOL (n=26 PAPERS)
Dahiya, R., Le, S., Ring, J.K. and Watson, K. (2022). Big data analytics and competitive advantage: the strategic role of firm-specific knowledge. Journal of Strategy and Management, 15(2), pp.175-193.
Elia, G., Raguseo, E., Solazzo, G., & Pigni, F. (2022). Strategic business value from big data analytics: An empirical analysis of the mediating effects of value creation mechanisms. Information & Management.
Günther, W.A., Mehrizi, M.H.R., Huysman, M., Deken, F., & Feldberg, F. (2022). Resourcing with data: Unpacking the process of creating data-driven value propositions. Journal of Strategic Information Systems.
Saenz, J., de Guinea, A.O., & Penalba-Aguirrezabalaga, C. (2022). Value creation through marketing data analytics: The distinct contribution of data analytics assets and capabilities to unit and firm performance. Information & Management.
Song, J., et al. (2022). The source of SMEs' competitive performance in COVID-19: Matching big data analytics capability to business models. Information Systems Frontiers.
Shi, Y., Cui, T., & Liu, F. (2022). Disciplined autonomy: How business analytics complements customer involvement for digital innovation. Journal of Strategic Information Systems.
Zhang, D., Pan, S.L., Yu, J., & Liu, W. (2022). Orchestrating big data analytics capability for sustainability: A study of air pollution management in China. Information & Management.
Jensen, M.H., Nielsen, P.A. and Persson, J.S. (2023). Benefits from Big Data Analytics Projects: A Critical System Heuristics Approach to Boundary Judgements. ECIS.
Oufan, S., De Luca, L., Morgan, R. and Delbridge, R. (2023). Realising Value from Big Data Technology Adoption: Understanding the Role of Organisational Capabilities in the Affordance Actualization Process. In Proceedings of the International Conference on Information Systems (Vol. 14). Association of Information Systems.
Tarn, D.D. and Wang, J. (2023). Can data analytics raise marketing agility? - A sense-and-respond perspective. Information & Management, 60(2), p.103743.
Tremblay, M.C., Kohli, R., & Rivero, C. (2023). Data is the new protein: How the Commonwealth of Virginia built digital resilience muscle and rebounded from opioid and COVID shocks. MIS Quarterly.
Yeh, Y.T., Eden, R., Fielt, E., Syed, R., Donovan, R., Eley, R. and Staib, A. (2023). Unveiling the value of big data analytics use: A digital hospital case study. ECIS.
Hopf, K., Joshi, M., Stelmaszak, M. and Shollo, A. (2024). Crafting Ever-Changing Data Products: Towards a Human-Centered Process Model of Data Work. ECIS.
León, O., de la Fuente, D., Fernandez-Vazquez, S. and Puente, J. (2024). Big data analytics capabilities, direct and mediating relationships with innovative and business performance. Journal of Management Analytics, pp.1-20.
DATA AS ALGORITHMIC INTELLIGENCE (n=35 PAPERS)
Newell, S., & Marabelli, M. (2015). Strategic opportunities (and challenges) of algorithmic decision-making: A call for action on the long-term societal effects of 'datification'. Journal of Strategic Information Systems.
Galliers, R. D., et al. (2017). Datification and its human, organizational and societal effects: The strategic opportunities and challenges of algorithmic decision-making. Journal of Strategic Information Systems.
Davenport, T. H., et al. (2018). Feature artificial intelligence for the real world. Harvard Business Review.
May, A., et al. (2020). Realizing Digital Innovation from Artificial Intelligence. ICIS.
Rinta-Kahila, T., et al. (2021). Algorithmic decision-making and system destructiveness: A case of automatic debt recovery. European Journal of Information Systems.
Marjanovic, O., Cecez-Kecmanovic, D., and Vidgen, R. (2021). Algorithmic pollution: Making the invisible visible. Journal of Information Technology.
Mikalef, P., and Gupta, M. (2021). Artificial intelligence capability: Conceptualization, measurement calibration, and empirical study on its impact on organizational creativity and firm performance. Information & Management.
Fügener, A., Grahl, J., Gupta, A., and Ketter, W. (2021). Will humans-in-the-loop become Borgs? Merits and pitfalls of working with AI. MIS Quarterly.
Lou & Wu, (2021). AI on drugs: Can artificial intelligence accelerate drug development? Evidence from a large-scale examination of bio-pharma firms. MIS Quarterly.
Van den Broek, E., Sergeeva, A., & Huysman, M. (2021). When the machine meets the expert: An ethnography of developing AI for hiring. MIS Quarterly.
Enholm, I.M., et al. (2021). Artificial intelligence and business value: A literature review. Information Systems Frontiers.
Gregory, R.W., Henfridsson, O., Kaganer, E. and Kyriakou, H. (2021). The role of artificial intelligence and data network effects for creating user value. Academy of Management Review.
Sebastian Raisch and Sebastian Krakowski (2021). Artificial intelligence and management: The automation–augmentation paradox. Academy of Management Review.
Berente, N., Gu, B., Recker, J., & Santhanam, R. (2021). Managing artificial intelligence. MIS Quarterly.
Teodorescu et al. (2021). Failures of fairness in automation require a deeper understanding of human–ML augmentation. MIS Quarterly.

(*continued on next page*)

(*continued*)

DATA AS A TOOL (n=26 PAPERS)
Sturm, T. et al. (2021). Coordinating human and machine learning for effective organizational learning. MIS Quarterly.
Cheng, X., Su, L., Luo, X., Benitez, J. and Cai, S. (2022). The good, the bad, and the ugly: Impact of analytics and artificial intelligence-enabled personal information collection on privacy and participation in ridesharing. European Journal of Information Systems, 31(3), pp.339-363.
Shollo, A., Hopf, K., Thiess, T. and Müller, O. (2022). Shifting ML value creation mechanisms: A process model of ML value creation. Journal of Strategic Information Systems.
Fanti, L., Guarascio, D. and Moggi, M. (2022). From Heron of Alexandria to Amazon's Alexa: A stylized history of AI and its impact on business models, organization, and work. Journal of Industrial and Business Economics.
Gregory, R. W., Henfridsson, O., Kaganer, E., and Kyriakou, H. (2022). Data network effects: Key conditions, shared data, and the data value duality. Academy of Management Review.
Jamwal, A.et al. (2022). Machine learning applications for sustainable manufacturing: A bibliometric-based review for future research. Journal of Enterprise Information Management.
Chatterjee, S., Chaudhuri, R., Kamble, S., Gupta, S., & Sivarajah, U. (2022). Adoption of artificial intelligence and cutting-edge technologies for production system sustainability: A moderator-mediation analysis. Information Systems Frontiers.
Rana, N.P., Chatterjee, S., Dwivedi, Y.K. and Akter, S. (2022). Understanding dark side of artificial intelligence (AI) integrated business analytics: assessing firm's operational inefficiency and competitiveness. European Journal of Information Systems, 31(3), pp.364-387.
Beese, Jannis and Fahse, Tobias (2023). Modern centaurs: How humans and AI systems interact in sales forecasting. ECIS.
Sagodi, A.; Dremel, C.; and van Giffen, B. (2023). Sensemaking in AI-based digital innovations: Insights from a manufacturing case study. ECIS.
Chatterjee, S., Chaudhuri, R., Kamble, S., Gupta, S. and Sivarajah, U. (2023). Adoption of artificial intelligence and cutting-edge technologies for production system sustainability: A moderator-mediation analysis. Information Systems Frontiers, 25(5), pp.1779-1794.
Chhina, S., Chadhar, M., Firmin, S. and Tatnall, A. (2023). Unlocking the Power of Blockchain: Driving Digital Business Transformation. ACIS.
Du, K., Li, G., Li, L. and Zhang, W. (2023). Unravelling the Business Value of Artificial Intelligence. In AMCIS.
Hoffmann, M. and Mehler, M., 2023. An Industry-Specific Investigation on Artificial Intelligence Adoption: The Cases of Financial Services and Manufacturing. PACIS.
Krakowski, S., Luger, J., & Raisch, S. (2023). Artificial intelligence and the changing sources of competitive advantage. Strategic Management Journal.
Fosso Wamba, S., Queiroz, M.M., Pappas, I.O. and Sullivan, Y. (2024). Artificial Intelligence Capability and Firm Performance: A Sustainable Development Perspective by the Mediating Role of Data-Driven Culture. Information Systems Frontiers, pp.1-15.
Stohr, A., Ollig, P., Keller, R. and Rieger, A. (2024). Generative mechanisms of AI implementation: A critical realist perspective on predictive maintenance. Information and Organization, 34(2), p.100503.

Note: 11 out of 142 papers are not listed under any of the types mentioned above. These papers are categorized as "Not Applicable" because they are either literature reviews or do not specify a particular purpose for data use.

References

Aaltonen, A., Alaimo, C., Kallinikos, J., 2021. The Making of Data Commodities: Data Analytics as an Embedded Process. J. Manag. Inf. Syst. 38, 401–429. https://doi.org/10.1080/07421222.2021.1912928.

Aaltonen, A., Alaimo, C., Parmiggiani, E., Stelmaszak, M., Jarvenpaa, S., 2023. What is Missing from Research on Data in Information Systems? Insights from the Inaugural Workshop on Data Research. Communications of the Association for Information Systems 53, pp-pp.

Abbasi, A., Sarker, S., Chiang, R.H.L., Lindner, C.H., 2016. Big Data Research in Information Systems: Toward an Inclusive Research Agenda. J Assoc Inf Syst.

Agarwal, A., Dahleh, M., Sarkar, T., 2019. A marketplace for data: An algorithmic solution, in: Proceedings of the 2019 ACM Conference on Economics and Computation. Association for Computing Machinery, Inc, pp. 701–726. doi: 10.1145/3328526.3329589.

Akhlaghpour, S., Wu, J., Lapointe, L., Pinsonneault, A., 2013. The ongoing quest for the IT artifact: Looking back, moving forward. J. Inf. Technol. 28, 150–166.

Alaimo, C., Kallinikos, J., 2022. Organizations Decentered: Data Objects, Technology and Knowledge. Organ. Sci. 33, 19–37. https://doi.org/10.1287/ORSC.2021.1552.

Alfaro, E., Bressan, M., Girardin, F., Murillo, J., Someh, I., Wixom, B.H., 2019. BBVA's Data Monetization Journey. MIS Q. Exec. 18, 111–128. https://doi.org/10.17705/2msqe.00011.

Alvesson, M., Kärreman, D., 2007. Constructing Mystery: Empirical Matters in Theory Development. Acad. Manag. Rev. 32, 1265–1281.

Anand, A., Sharma, R., Coltman, T., 2016. Four Steps to Realizing Business Value from Digital Data Streams. MIS Q. Exec.

Aral, S., Weill, P., 2007. IT assets, organizational capabilities, and firm performance: How resource allocations and organizational differences explain performance variation. Organ. Sci. 18, 763–780. https://doi.org/10.1287/orsc.1070.0306.

Baesens, B., Bapna, R., Marsden, J.R., Vanthienen, J., Zhao, J.L., 2014. Transformational issues of big data and analytics in networked business. MIS Q.

Baird, A., Maruping, L.M., 2021. The next generation of research on is use: A theoretical framework of delegation to and from agentic is artifacts. MIS Q. 45, 315–341. https://doi.org/10.25300/MISQ/2021/15882.

Barkin, S.R., Dickson, G.W., 1977. An Investigation on Information System Utilization.

Bekmamedova, N., Shanks, G., 2014. Social media analytics and business value: A theoretical framework and case study, in: Proceedings of the Annual Hawaii International Conference on System Sciences. IEEE Computer Society, pp. 3728–3737. doi: 10.1109/HICSS.2014.464.

Benbasat, I., Zmud, R.W., 2003. The Identity Crisis within the Is Discipline: Defining and Communicating the Discipline's Core Properties. MIS Q. 27, 183–194.
Berente, N., Gu, B., Recker, J., Santhanam, R., 2021. Managing Artificial Intelligence. MIS Q. https://doi.org/10.25300/MISQ/2021/16274.
Birch, K., Cochrane, D.T., Ward, C., 2021. Data as asset? The measurement, governance, and valuation of digital personal data by Big Tech. Big Data Soc 8. https://doi.org/10.1177/20539517211017308.
Blohm, I., Marco Leimeister, J., Krcmar, H., 2013. Crowdsourcing: How to Benefit from (Too) Many Great Ideas. MIS Q. Exec. 12.
Breslin, D., Gatrell, C., 2023. Theorizing through literature reviews: The miner-prospector continuum. Organ. Res. Methods 26 (1), 139–167.
Brynjolfsson, E., McElheran, K., 2016. The rapid adoption of data-driven decision-making, in. American Economic Review. American Economic Association 133–139. https://doi.org/10.1257/aer.p20161016.
Brynjolfsson, E., Mitchell, T., 2017. What can machine learning do? Workforce implications 2017. Science.
Castelvecchi, D., 2016. Can we open the black box of AI? Nature News.
Chen, H., Chiang, R.H.L., Storey, V.C., Lindner, C.H., Robinson, J.M., 2012. Business Intelligence and Analytics: From Big Data to Big Impact. MIS Q. 36, 1165–1188.
Chen, Y.J., Huang, K.W., 2016. Pricing data services: Pricing by minutes, by gigs, or by megabytes per second? Inf. Syst. Res. 27, 596–617. https://doi.org/10.1287/isre.2016.0651.
Chen, D.Q., Preston, D.S., Swink, M., 2015. How the use of big data analytics affects value creation in supply chain management. J. Manag. Inf. Syst. 32, 4–39. https://doi.org/10.1080/07421222.2015.1138364.
Constantiou, I.D., Kallinikos, J., 2015. New games, new rules: Big data and the changing context of strategy. J. Inf. Technol. 30, 44–57. https://doi.org/10.1057/jit.2014.17.
Cornelissen, J., 2017. Editor's comments: Developing propositions, a process model, or a typology? Addressing the challenges of writing theory without a boilerplate. Acad. Manag. Rev. 42, 1–9. https://doi.org/10.5465/amr.2016.0196.
Côrte-Real, N., Oliveira, T., Ruivo, P., 2017. Assessing business value of Big Data Analytics in European firms. J. Bus. Res. 70, 379–390.
Cronin, M.A., George, E., 2023. The why and how of the integrative review. Organ. Res. Methods 26 (1), 168–192.
Davenport, T.H., Ronanki, R., Wheaton, J., Nguyen, A., 2018. Feature Artificial Intelligence for the Real World. Harv. Bus. Rev.
DeLone, W.H., McLean, E.R., 2003. The DeLone and McLean model of information systems success: A ten-year update, in: Journal of Management Information Systems. M.E. Sharpe Inc., pp. 9–30. doi: 10.1080/07421222.2003.11045748.
Delone, W.H., Mclean, E.R., 1992. Information Systems Success: The Quest for the Dependent Variable. Research 3, 60–95.
Devaraj, S., Kohli, R., 2003. Performance Impacts of Information Technology: Is Actual Usage the Missing Link? Manage Sci 49, 273–289.
Dick, S., 2019. Artificial Intelligence. Harvard Data Science Review. https://doi.org/10.1162/99608f92.92fe150c.
Dosi, G., 2000. The Nature and Dynamics of Organizational Capabilities. Oxford University Press.
Drucker, P.F., 1999. Knowledge-Worker Productivity: The Biggest Challenge. Calif. Manage. Rev.
Duan, Y., Cao, G., Edwards, J.S., 2020. Understanding the impact of business analytics on innovation. Eur. J. Oper. Res. 281, 673–686. https://doi.org/10.1016/j.ejor.2018.06.021.
Ekbia, H.R., 2009. Digital artifacts as quasi-objects: Qualification, mediation, and materiality. J. Am. Soc. Inf. Sci. Technol.
Enholm, I.M., Papagiannidis, E., Mikalef, P., Krogstie, J., 2021. Artificial Intelligence and Business Value: A Literature Review. Inf. Syst. Front. https://doi.org/10.1007/s10796-021-10186-w.
Essén, A., Knudsen, M., Alvesson, M., 2022. Explaining ignoring: Working with information that nobody uses. Organ. Stud. 43 (5), 725–747.
Farronato, C., Maccormack, A., Researcher, C., Mehta, S., 2020. Innovation at Uber: The Launch of Express POOL. Available at: Harv. Bus. Rev. www.hbsp.harvard.edu.
Faulkner, P., Runde, J., 2019. Theorizing the digital object. MIS Q. 43, 1278–1302. https://doi.org/10.25300/MISQ/2019/13136.
Feldman, M.S., March, J.G., 1981. Information in organizations as signal and symbol. Adm. Sci. Q. 171–186.
Galliers, R.D., Newell, S., Shanks, G., Topi, H., 2017. Datification and its human, organizational and societal effects: The strategic opportunities and challenges of algorithmic decision-making. J. Strateg. Inf. Syst. 26, 185–190. https://doi.org/10.1016/j.jsis.2017.08.002.
Ghasemaghaei, M., Calic, G., 2020. Assessing the impact of big data on firm innovation performance: Big data is not always better data. J. Bus. Res. 108, 147–162.
Gherardi, S., 2000. Practice based theorizing on learning and knowing in organizations. Orgnization 7.
Gomez-Uribe, C.A., Hunt, N., 2015. The Netflix recommender system: Algorithms, business value, and innovation. ACM Trans. Manag. Inf. Syst. 6. https://doi.org/10.1145/2843948.
Gregor, S., 2006. The Nature of Theory in Information Systems. MIS Q. 30, 611–642.
Gregory, R.W., Henfridsson, O., Kaganer, E., Kyriakou, S.H., 2021. The role of artificial intelligence and data network effects for creating user value. Acad. Manag. Rev. 46, 534–551. https://doi.org/10.5465/amr.2019.0178.
Grover, V., Chiang, R.H.L., Liang, T.P., Zhang, D., 2018. Creating Strategic Business Value from Big Data Analytics: A Research Framework. J. Manag. Inf. Syst. 35, 388–423. https://doi.org/10.1080/07421222.2018.1451951.
Günther, W.A., Mehrizi, M.H.R., Huysman, M., Deken, F., & Feldberg, F., 2022. Resourcing with data: Unpacking the process of creating data-driven value propositions. Journal of Strategic Information Systems.
Günther, W.A., Rezazade Mehrizi, M.H., Huysman, M., Feldberg, F., 2017. Debating big data: A literature review on realizing value from big data. J. Strateg. Inf. Syst. 26, 191–209. https://doi.org/10.1016/j.jsis.2017.07.003.
Hagiu, A., Wright, J., 2020. Data-enabled learning, network effects and competitive advantage *. RAND J. Econ.
Jones, M., 2019. What we talk about when we talk about (big) data. J. Strateg. Inf. Syst. 28, 3–16. https://doi.org/10.1016/j.jsis.2018.10.005.
Jones, C.I., Tonetti, C., 2020. Nonrivalry and the Economics of Data. Am. Econ. Rev. 110, 2819–2858. https://doi.org/10.1257/aer.20191330.
Kallinikos, J., Aaltonen, A., Marton, A., 2013. The Ambivalent Ontology of Digital Artifacts. MIS Q. 37, 357–370.
Karanasios, S., Nardi, B., Spinuzzi, C., Malaurent, J., 2021. Moving forward with activity theory in a digital world. Mind Cult. Act. 28, 234–253. https://doi.org/10.1080/10749039.2021.1914662.
Kellogg, K.C., Valentine, M.A., Christin, A., 2020. Algorithms at work: The new contested terrain of control. Acad. Manag. Ann. 14, 366–410. https://doi.org/10.5465/annals.2018.0174.
Knox, K.T., 2007. The Various and Conflicting Notions of Information, Issues in Informing Science and Information Technology.
Krakowski, S., Luger, J., Raisch, S., 2023. Artificial intelligence and the changing sources of competitive advantage. Strateg. Manag. J. 44, 1425–1452. https://doi.org/10.1002/smj.3387.
Krishnamoorthi, S., Mathew, S.K., 2018. Business analytics and business value: A comparative case study. Inf. Manag. 55, 643–666. https://doi.org/10.1016/j.im.2018.01.005.
Labadie et al, 2020. FAIR Enough? Enhancing the Usage of Enterprise Data with Data Catalogs. IEEE 22nd Conference on Business Informatics (CBI).
Lange, H.E., Drews, P., Höft, M., 2021. Realization of Data-Driven Business Models in Incumbent Companies: An Exploratory Study Based on the Resource-Based View, in: ICIS.
Lefebvre, H., Legner, C., Fadler, M., 2021. Data democratization: toward a deeper understanding. ICIS.
Leidner, D.E., 2018. Review and theory symbiosis: An introspective retrospective. J Assoc Inf Syst 19, 552–567. https://doi.org/10.17705/1jais.00501.
Matook, S., Brown, S.A., 2017. Characteristics of IT artifacts: a systems thinking-based framework for delineating and theorizing I. Inf. Syst. J.
Mehta, S., Dawande, M., Janakiraman, G., Mookerjee, V., 2021. How to Sell a Data Set? Pricing Policies for Data Monetization. Inf. Syst. Res. 32, 1281–1297. https://doi.org/10.1287/isre.2021.1027.
Miettinen, R., Samra-Fredericks, D., Yanow, D., 2009. Re-turn to practice: An introductory essay. Organ. Stud. 30, 1309–1327. https://doi.org/10.1177/0170840609349860.
Mikalef, P., Boura, M., Lekakos, G., Krogstie, J., 2019. Big data analytics and firm performance: Findings from a mixed-method approach. J. Bus. Res. 98, 261–276. https://doi.org/10.1016/j.jbusres.2019.01.044.

Möhlmann, M., Zalmanson, L., Henfridsson, O., Gregory, R.W., 2021. Algorithmic Management of Work on Online Labor Platforms: When Matching Meets Control. MIS Q. 45, 1999–2022. https://doi.org/10.25300/MISQ/2021/15333.
Murray, A., Rhymer, J.E.N., Sirmon, D.G., 2021. Humans and technology: Forms of conjoined agency in organizations. Acad. Manag. Rev. 46 (3), 552–571.
Najjar, M.S., Kettinger, W.J., 2013. Data Monetization: Lessons from a Retailer's Journey. MIS Q. Exec.
Newell, S., Marabelli, M., 2015. Strategic opportunities (and challenges) of algorithmic decision-making: A call for action on the long-term societal effects of "datification". J. Strateg. Inf. Syst. 24, 3–14. https://doi.org/10.1016/j.jsis.2015.02.001.
Olszak, C.M., Zurada, J., 2020. Big Data in Capturing Business Value. Inf. Syst. Manag. 37, 240–254. https://doi.org/10.1080/10580530.2020.1696551.
Orlikowski, W., Iacono, S., 2001. Desperately seeking the "IT" in IT research - A call to Theorizing the IT Artifact. Inf. Syst. Res.
Paré, G., Trudel, M.C., Jaana, M., Kitsiou, S., 2015. Synthesizing information systems knowledge: A typology of literature reviews. Inf. Manag. 52, 183–199. https://doi.org/10.1016/j.im.2014.08.008.
Parvinen, P., Pöyry, E., Gustafsson, R., Laitila, M., Rossi, M., 2020. Advancing data monetization and the creation of data-based business models. Commun. Assoc. Inf. Syst. 47, 25–49. https://doi.org/10.17705/1CAIS.04702.
Piccoli, G., Rodriguez, J., Someh, I.A., Wixom, B., 2022. Data Liquidity: Conceptualization. Measurement and Determinants, ICIS.
Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., 2019. Language Models are Unsupervised Multitask Learners. OpenAI Blog.
Raisch, S., Krakowski, S., 2021. Artificial intelligence and management: The automation–augmentation paradox. Acad. Manag. Rev. 46, 192–210. https://doi.org/10.5465/AMR.2018.0072.
Rana, N.P., Chatterjee, S., Dwivedi, Y.K., Akter, S., 2022. Understanding dark side of artificial intelligence (AI) integrated business analytics: assessing firm's operational inefficiency and competitiveness. Eur. J. Inf. Syst. 31 (3), 364–387.
Ray, J., Menon, S., Mookerjee, V., 2020. Bargaining over Data: When Does Making the Buyer More Informed Help? Inf. Syst. Res. 31, 1–15. https://doi.org/10.1287/ISRE.2019.0872.
Rinta-Kahila, T., Someh, I., Gillespie, N., Indulska, M., Gregor, S., 2021. Algorithmic decision-making and system destructiveness: A case of automatic debt recovery. Eur. J. Inf. Syst. https://doi.org/10.1080/0960085X.2021.1960905.
Rowley, J., 2007. The wisdom hierarchy: representations of the DIKW hierarchy. J. Inf. Sci. 33 (2), 163–180.
Sadiq, S., Aryani, A., Demartini, G., Hua, W., Indulska, M., Burton-Jones, A., Khosravi, H., Benavides-Prado, D., Sellis, T., Someh, I., Vaithianathan, R., Wang, S., Zhou, X., 2022. Information Resilience: the nexus of responsible and agile approaches to information use, in: VLDB Journal. Springer Science and Business Media Deutschland GmbH. doi: 10.1007/s00778-021-00720-2.
Sandberg, J., Alvesson, M., 2021. Meanings of Theory: Clarifying Theory through Typification. J. Manag. Stud. 58, 487–516. https://doi.org/10.1111/joms.12587.
Schilling, M.A., 2000. Toward a general modular systems theory and its application to interfirm product modularity. Acad. Manag. Rev.
Schryen, G., 2013. Revisiting IS business value research: what we already know, what we still need to know, and how we can get there. Eur. J. Inf. Syst. 22, 139–169. https://doi.org/10.1057/ejis.2012.45.
Schryen, G., Wagner, G., Benlian, A., Paré, G., 2020. A knowledge development perspective on literature reviews: Validation of a new typology in the IS field. Commun. Assoc. Inf. Syst. 46, 134–186. https://doi.org/10.17705/1CAIS.04607.
Schuetz, S.W., Venkatesh, V., 2020. The Rise of Human Machines: How Cognitive Computing Systems Challenge Assumptions of User-System Interaction. Journal of the AIS 21, 2020.
Schüritz, R., Seebacher, S., Dorner, R., 2017. Capturing Value from Data: Revenue Models for Data-Driven Services, in: HICSS.
Seidel, S., Berente, N., 2020. Automate, informate, and generate: affordance primitives of smart devices and the internet of things., in: Handbook of Digital Innovation.
Shamim, S., Zeng, J., Shariq, S.M., Khan, Z., 2019. Role of big data management in enhancing big data decision-making capability and quality among Chinese firms: A dynamic capabilities view. Inf. Manag. 56. https://doi.org/10.1016/j.im.2018.12.003.
Shollo, A., Galliers, R.D., 2016. Towards an understanding of the role of business intelligence systems in organisational knowing. Inf. Syst. J. 26, 339–367. https://doi.org/10.1111/isj.12071.
Shollo, A., Hopf, K., Thiess, T., Müller, O., 2022. Shifting ML value creation mechanisms: A process model of ML value creation. J. Strateg. Inf. Syst. 31, 101734. https://doi.org/10.1016/j.jsis.2022.101734.
Spiekermann, M., 2019. Data Marketplaces: Trends and Monetisation of Data Goods. Intereconomics 54, 208–216. https://doi.org/10.1007/s10272-019-0826-z.
Surbakti, F.P.S., Wang, W., Indulska, M., Sadiq, S., 2020. Factors influencing effective use of big data: A research framework. Inf. Manag. 57. https://doi.org/10.1016/j.im.2019.02.001.
Teodorescu, M.H.M., Morse, L., Awwad, Y., Kane, G.C., 2021. Failures of fairness in automation require a deeper understanding of human–ml augmentation. MIS Q. 45, 1483–1499. https://doi.org/10.25300/MISQ/2021/16535.
Thi Ngoc, N., Elbanna, A., 2023. Understanding human-AI augmentation in business and management context. ECIS. 5–11.
Trieu, V.H., 2017. Getting value from Business Intelligence systems: A review and research agenda. Decis. Support Syst. 93, 111–124.
Wade, M., Hulland, J., 2004. Review: The resource-based view and information systems research: Review, extension, and suggestions for future research. MIS Q. 28, 107–142. https://doi.org/10.2307/25148626.
Wang, Y., Byrd, T.A., 2017. Business analytics-enabled decision-making effectiveness through knowledge absorptive capacity in health care. J. Knowl. Manag. 21, 517–539. https://doi.org/10.1108/JKM-08-2015-0301.
Watson, R.T., Webster, J., 2020. Analysing the past to prepare for the future: Writing a literature review a roadmap for release 2.0. J. Decis. Syst. 29, 129–147. https://doi.org/10.1080/12460125.2020.1798591.
Webster, J., Watson, R.T., 2002. Analyzing the Past to Prepare for the Future: Writing a Literature Review. MIS Quarterly 26, xiii–xxiii.
Wixom, B.H., Farrell, K., 2019. Building Data Monetization Capabilities that Pay Off. MIT CISR briefing.
Wixom, B.H., Piccoli, G., Rodriguez, J., 2021. Fast-Track Data Monetization With Strategic Data Assets. MIT Sloan Manag Rev.
Wixom, B.H., Ross, J.W., 2017. How to Monetize Your Data. MIT Sloan Manag. Rev.
Xu, D., Someh, I.A., Indulska, M., 2023. Towards a reconceptualization of data in organizations: a literature review. ECIS.
Ye, H., Yang, X., Wang, X., Stratopoulos, T.C., 2021. Monetization Of Digital Content: Drivers Of Revenue On Q&A Platforms. J. Manag. Inf. Syst. 38, 457–483. https://doi.org/10.1080/07421222.2021.1912934.
Yuchtman, E., Seashore, S.E., 1967. A System Resource Approach to Organizational Effectiveness. Source, American Sociological Review.
Zins, C., 2007. Conceptual approaches for defining data, information, and knowledge. J. Am. Soc. Inf. Sci. Technol. 58, 479–493. https://doi.org/10.1002/asi.20508.
Zolnowski, R., 2016. Business Model Transformation Patterns of Data-Driven Innovations. ECIS.
Zuboff, S., 1985. Automate/Informate: The two faces of intelligent technology. Organizational Dynamics.